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RESEARCH

An experimentally validated magnetic force model for discrete element modeling of paramagnetic granular media

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Abstract
Magnetic interactions between metallic granular particles can lead to magnetic cohesion, influencing the flow characteristics of granular media. This magnetic cohesion has been studied in the context of Magneto-Rheological Fluids (MRFs) for their unique flow properties and their use across multiple industries. In Planetary Science, magnetic cohesion can influence regolith behavior on metallic asteroids with remnant magnetic fields. The upcoming NASA Psyche mission will study the metallic asteroid 16-Psyche, which is expected to have a surface magnetic field. Modeling and simulating the effect of magnetic cohesion on granular media is crucial for accurately simulating the behavior of magnetic granular materials in both terrestrial and planetary applications. We introduce an improved magnetic force model in LIGGGHTS, an open-source discrete element modeling software, to calculate magnetic forces between paramagnetic grains. The model is based on the Mutual Dipole Method and the Inclusion Model, extensions of the Fixed Dipole Method. We validate the model using 1-D unit tests and compare the results of avalanche simulations of paramagnetic regolith with experimental data. This work contributes to understanding the role of magnetic cohesion in small body surface processes and provides a tool for future studies of magnetic granular materials in DEM.

Keywords Magnetic cohesion · Magnetic dipoles · Multipole methods · Magnetic force model · MRF · Asteroid psyche · SSDEM · Numerical modeling

1 Introduction

Magnetic interactions between metallic granular particles can result in magnetic cohesion [1]. Various theoretical, experimental, and numerical studies have looked at the effect of magnetic cohesion on the angle of repose and flow characteristics of granular media for Magneto-Rheological Fluids (MRF)[2–6] and Planetary Science [7, 8] applications. Forsyth et al. [9] conducted experiments with steel ball bearings to show that the angle of repose changes linearly but slowly with increasing magnetic field strength. Fazekas et al. [10] & Taylor et al. [11] performed Discrete Element Modeling (DEM) simulations to validate the experimental results of [9] and provide varying reasons for a slight increase in the angle of repose, even for large cohesive forces. Peters and Lemaire [2] argued that, rather than a uniform magnetic field that results in a highly anisotropic force between particles, a rotating magnetic field is more suitable for control over the avalanching of cohesive particles.

Sunday et al. [8] conducted avalanching experiments with varied particle sizes and two materials to show transitions between three different flow regimes: the granular regime, where particles move independently, the correlated regime, where particles move in clusters, and the plastic regime, where the entire assembly acts as a single block. They showed that the linear increase of the angle of repose with applied magnetic field only holds in the granular regime, while surface roughness parameters: Root-Mean-Square (RMS) height and RMS slope continue to increase in the correlated regime.

Discrete Element Model (DEM) based numerical studies are used to model forces on each particle to simulate magnetic interactions between granular particles using various

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methods. Fazekas et al. [10] & Taylor et al. [11] used the Fixed Dipole Model (FDM) based on single particle dipoles for magnetic interactions between granular particles. However, various studies [3, 7, 12] have shown that dipolar models, such as FDM, are quite inaccurate when particles are sufficiently close, especially in dense assemblies. In this work, we seek to implement a more accurate magnetic force model for DEM simulations. We build upon the work of Keaveny and Maxey [3] who proposed the Inclusion Model. We present the first experimentally validated DEM model of magnetic interactions of granular materials.

1.1 Planetary science motivation

Due to surface gravities up to 1 million times smaller than that on Earth, surface grains on traditional (non-metallic) asteroids are significantly affected by van der Waals cohesion [13]. Multiple studies considered the role of cohesion in simulations related to NASA's successful Double Asteroid Redirection Test (DART) Mission. DART impacted the smaller satellite object Dimorphos in the binary pair of 65803 Didymos and Dimorphos to demonstrate the potential to change an asteroid's trajectory for planetary defense. Zhang et al. [14] studied the role of cohesion in the bulk structural stability to maintain the fast rotational spin of the primary asteroid Didymos. Raducan et al. [15] evaluated different values for momentum change for the impact on the satellite Dimorphos to show that Dimorphos must have a weaker cohesive strength than the larger parent Didymos. The OSIRIS-REx mission experienced unexpected behavior due to near-zero cohesion [16] during the sample collection on asteroid 101955 Bennu. Sánchez and Scheeres [17] looked at the cohesive properties of different fine powders in the context of rubble-pile asteroids. These studies demonstrate the need to model and simulate cohesion for asteroid regolith research.

NASA's Psyche spacecraft, launched in October 2023, is headed to the main-belt asteroid 16-Psyche. A predominantly metallic body, 16-Psyche, is assumed to be a remnant core of a protoplanet [18]. A dynamo could have existed in the initial cooling period of the asteroid [19]. This dynamo would result in a magnetic field being locked in the body. Anand et al. [20] model possible magnetic field of 16-Psyche due to Solar Wind Induced Magnetization (WIM). Courville et al. [21] theorizes volcanic dykes on 16-Psyche with magnetic field signatures. Solar plasma wind interaction simulations by Fatemi and Poppe [22] assume surface magnetic fields up to 500 nT. These various theories suggest the possibility of a local surface magnetic field on 16-Psyche. The spacecraft is carrying a magnetometer [23, 24] to measure the magnetic field from orbit.

The likely constituents of M-type asteroids are iron and nickel. These ferromagnetic substances will have paramagnetic ores, resulting in a paramagnetic regolith. Kohout et al. [25] measured magnetic susceptibilities of multiple iron and stony-iron meteorites. When placed in a uniform magnetic field, paramagnetic grains develop a magnetic moment. This moment results in a magnetic force acting between the particles, increasing the net cohesion. Sikka et al. [7] argued that on metallic bodies like 16-Psyche, magnetic cohesion can dominate over surface gravity for sub-micron-sized grains, while noting that a higher magnetic field strength would further enhance this effect for larger particles.

Robinson et al. [26] showed different gravity-driven morphological features on Eros 433's three craters: Psyche, Himeros, and Shoemaker. Avalanching on the Moon was compared to the same on Earth by Kokelaar et al. [27]. They showed that Lunar avalanches only involve dry-flowing grains, while terrestrial avalanches involve both dry and fluidized grains. Sugita et al. [28] demonstrated evidence of mass-wasting on asteroid 162173 Ryugu using Hayabusa2 observations. The role of volatiles in mass wasting on Vesta and Ceres was studied by Parekh et al. [29]. They concluded that dry granular flows dominate Vesta's mass-wasting, while Ceres shows evidence of fluidized flows. Tang et al. [30] studied mass-wasting on Bennu's Bralgah crater using DEM simulations. All these studies have shown the importance of cohesion in understanding mass wasting and granular flows on small planetary bodies.

Based on granular mechanics, DEM simulations have been used to model the regolith on small bodies. Researchers have used DEM in multiple studies for rubble pile asteroids ranging from reshaping & disruption [31], avalanching [32], and wave transmission [33]. Maurel et al. [34] simulated the Brazil-nut effect on asteroid regolith. DeMartini et al. [35] used a proprietary software PKDGRAV to study and simulate the 2029 tidal encounter of Apophis 99942 with Earth. Frizzell and Hartzell [36–38], simulate wave transmission on regolith to study the cold spot formation around lunar craters. Marohnic et al. [39] looked at the effect of particle shape on rubble-pile dynamics with a new approach to model non-spherical particles. Tang et al. [30] uses the PKDGRAV code to study mass-wasting on the asteroid 101955 Bennu's Bralgah Crater. Many researchers studied the various aspects of the DART mission target: the Dimorphos and Didymos system, including the impact momentum enhancement factor using an integrated smoothed particle hydrodynamics (SPH) and DEM code [40] and the effect of cohesion on the structural stability of Didymos using PKDGRAV [14].

1.2 Magneto-rheological fluids motivation

Magneto-Rheological Fluids (MRF) are suspensions of micron-sized, magnetizable particles in a carrier fluid [41]. Without an external magnetic field, the particles are randomly distributed in the fluid, and the fluid behaves like a Newtonian fluid. When subjected to an applied magnetic field, the particles become magnetized and form chain-like structures along the field lines, thereby altering the bulk rheological properties, such as viscosity. This property of MRFs is exploited in various applications, including dampers, clutches, and prosthetics. Bica et al. [42] provide a comprehensive review of MRF applications in various industries. Understanding the behavior of MRF in various magnetic environments is crucial for optimizing these applications.

Modeling and simulation of MRFs is crucial for predicting their behavior under various conditions. There are two primary approaches to modeling MRFs: continuum models and discrete models. Ghaffari et al. [43] provide a review of both the modeling and simulation techniques used for MRFs. Continuum models treat the system as a continuous medium modeled with rheological equations that describe fluid flow, magnetization, structural formation, and stress–strain relationships. These models are computationally faster but are limited by the rheological equation used and cannot capture particle-level interactions for every application. Finding an accurate rheological equation for MRFs is an ongoing area of research. Discrete models, on the other hand, simulate the behavior of magnetized particles independently of the carrier fluid, enabling a more detailed understanding of particle interactions and dynamics. While these models can be computationally expensive, they provide valuable insights into the microstructural behavior of MRFs. Modeling interparticle magnetic forces is a key research area for discrete models. Various methods have been proposed to model magnetic interactions between particles in discrete models.

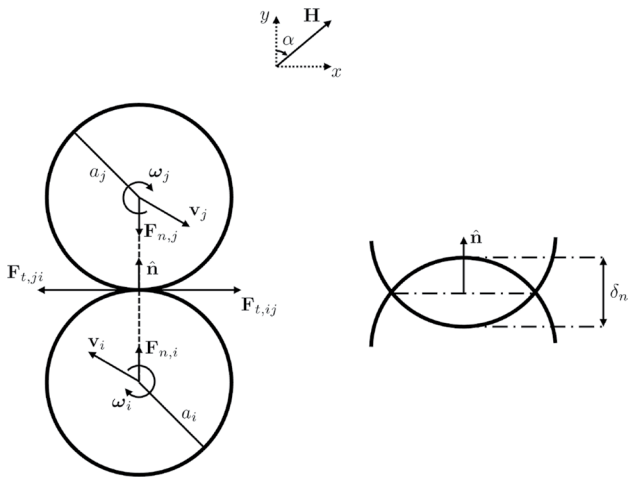


Fig. 1 Soft-sphere discrete element model. a represents the particle radius, $\hat{n}$ represents the normal at the point of contact, H is the applied magnetic field, v and $\boldsymbol{\omega}$ are the particle velocity and angular velocity respectively, and F_t and F_n are the tangential and normal forces on each particle. Subscripts i and j represent the two particles. δ_n is the overall distance between the particles during collision. The applied magnetic field is at an angle α to the line joining the two particles. Velocities and forces are not necessarily in the same plane as the magnetic field

Han et al. [4] reviewed different magnetic force models used in DEM simulations of MRFs. Most commonly, dipolar models like the Fixed Dipole Method (FDM) and the Mutual Dipole Method (MDM) are employed due to their simplicity and computational efficiency. However, these models have limitations in accuracy, especially when particles are in close proximity. Keaveny and Maxey [3] proposed the Inclusion Model, which improves upon dipolar models, but Han et al. [4] & Ghaffari et al. [43] note that it is highly computationally expensive and therefore not widely adopted. We introduce a critical magnetic time step that allows the Inclusion Model to be run at a larger time step than the Hertzian collision time, making it more computationally feasible for DEM simulations. Our model is implemented in LIGGGHTS, an open-source DEM software, and validated against experiments. This work is the first experimentally validated Inclusion model for magnetic interactions in DEM simulations.

1.3 Soft-sphere discrete element model

Our simulations utilize a Soft-Sphere Discrete Element Modeling (SSDEM) technique to model the interactions between thousands of grains. In discrete element models, equations of motion are integrated at each time step for every particle in the system. Soft-sphere, in particular, means that particles themselves can deform during collisions. Various force models are used to model the different physics of the problem, such as a spring-dashpot for collision dynamics.

Consider a system with a total of N particles. The dynamical equations for a the ith particle (Fig. 1) are given as

$$m_i \dot{\mathrm{v}}_i = m_i \mathrm{g} + \sum_{j=1}^{N} (\mathrm{F}_{n,i} + \mathrm{F}_{c,i} + \mathrm{F}_{t,i}) \ , \tag{1}$$

$$I_i \dot{\boldsymbol{\omega}}_i = \sum_{j=1}^{N} (\mathrm{T}_i - \mathrm{M}_{f,i}) \ , \tag{2}$$

where m_i, I_i, v_i, $\boldsymbol{\omega}_i$ are mass, moment of inertia, translational and rotational velocity respectively of ith particle. $\mathrm{F}_{n,i}$, $\mathrm{F}_{c,i}$, $\mathrm{F}_{t,i}$ are normal, cohesive and tangential forces respectively on the ith particle. Finally, T_i and $\mathrm{M}_{f,i}$ are the

torque and frictional moment on i^{th} particle. The torque is given by $\mathrm{T}_i = a_i \mathrm{F}_{t,i} \times \hat{\mathrm{n}}$ where a_i is the radius of the particle and $\hat{\mathrm{n}}$ is the direction of the normal as shown in the Fig. 1.

For collision dynamics, we use the Hertz contact model, where the contact force experienced by the particles depends on the overlap 'δ' (Fig. 1) as a power law $\delta^{3/2}$. The Hertz contact model requires the simulation time step (*dt*) to be less than the Hertzian collision time (τ_{hc}) [44], that is usually in 10^{-6} order of magnitude. Along with the contact model, we use an elastic spring and dashpot-based static and rolling friction model [45]. The cohesion is a simplified JKR - Johnson-Kendall-Roberts (SJKR) model [46] based on cohesion energy density. The cohesive force is given by $\mathrm{F}_c = kA\hat{\mathrm{n}}$ where *k* is the cohesion energy density in J/m^3 units, and A is the contact area between the two particles.

1.4 Paper overview

This paper presents the implementation of the modified Inclusion model for magnetic interactions of spherical particles, as proposed by Keaveny and Maxey [3], implemented in the open-source DEM software LIGGGHTS [47] and validated with experimental results from Sunday et al. [8]. Section 2 reviews the magnetostatics equations and the theoretical background for magnetic force models for granular particles.

Section 3 describes the implementation of the magnetic force model in the open-source DEM software LIGGGHTS. We detail the numerical methods, including the Mutual Dipole Method (MDM) and the Inclusion Model, and discuss computational considerations for efficient simulation. We also introduce the critical magnetic time step, which allows the magnetic force model to be run at a larger time step than the Hertzian collision time.

Section 4 presents validation of the implemented models against analytical and published results for simple particle systems, demonstrating the accuracy and limitations of each approach.

Section 5 applies the model to simulate avalanching of metallic grains under varying magnetic field strengths, comparing simulation outcomes to experimental results. We replicate flow regimes along with trends for angle of repose, surface roughness, and root mean square (RMS) slope, demonstrating the model's effectiveness in capturing magnetic cohesion effects.

Section 7 concludes with a summary of contributions, limitations, and directions for future work.

2 Magnetostatics

Consider a system of *N* spherical particles placed in a uniform magnetic field H_0. Each particle has a radius *a* and a magnetic susceptibility χ. Neglecting free and displacement currents, we can write Laplace's equations [48] using scalar magnetic potential Φ such that $\mathrm{H} = -\nabla\Phi$:

$$\nabla^2\phi_{out} = 0 \quad \nabla^2\phi_{in,i} = 0 \ , \tag{3}$$

where ϕ_{out} is the potential outside the particle and $\phi_{in,i}$ is the potential inside the ith particle. The boundary conditions for the Laplace equations at particle surfaces ($r_n = a$) are given by:

$$\phi_{out} = \phi_{in,i} \quad \mu\frac{\partial\phi_{in,i}}{\partial r_n} = \mu_0\frac{\partial\phi_{out}}{\partial r_n} \ , \tag{4}$$

where μ is the permeability of the particle and μ_0 is the permeability of free space. Additionally, as $r_n \to \infty$, $\phi_{out} \to -\mathrm{H}_0 \cdot \mathbf{r_n}$.

The general solution to the Laplace equations (Eq. 3) is an infinite sum of spherical harmonics. Magnetic force models are based on an approximation of this infinite series.

Earlier studies [10, 11] employed the Fixed Dipole Method (FDM) to model magnetic interactions between granular particles. Recent studies [4–6, 49] pivot to the more accurate Mutual Dipole Model (MDM). Sikka et al. [7] compared different models of magnetic force, such as the Mutual Dipole Method (MDM) [3], the Finite Volume Model (FVM) [12], and the Spherical Harmonic Approximation (SHA) [3]. They concluded that SHA presents the most accurate results while being fairly computationally efficient. FVM can provide comparable accuracy to the SHA, but it is significantly slower. While being fast, the MDM loses accuracy when particles are closer than 3 particle radii. However, the SHA only applies to a two-particle system and, therefore, cannot be applied to a system of multiple particles. Keaveny and Maxey [3] provided an Inclusion model to calculate the force on each particle placed in a uniform magnetic field by combining dipole-based models such as FDM and MDM, with SHA, thereby providing a potential framework for multi-particle systems. However, to the best of our knowledge, the Inclusion model has not been implemented in any DEM software. We specifically implemented and compared the Inclusion model to the MDM.

2.1 Mutual Dipole Method

The Mutual Dipole Model (MDM) extends the Fixed Dipole Method (FDM), the simplest model. In both cases, the magnetic moment of a particle is assumed to be dipolar. The FDM only uses the external magnetic field to calculate the magnetization for each particle. A single particle placed in the uniform magnetic field H_0 gains a dipole moment,

$$\mathrm{m} = C\mathrm{H}_0, \qquad C = \frac{4}{3}\pi a^3 \chi_{eff} \ , \tag{5}$$

where a is the particle radius and $\chi_{eff} = \frac{3\chi}{\chi+3}$ is particle effective susceptibility.

FDM assumes that every particle is dipolar and doesn't affect the others. MDM addresses the second assumption by incorporating the effect of the magnetic field generated by every other particle. A particle with dipole moment m changes the magnetic field around it by H_{dip} [3],

$$\mathrm{H}_{dip}(\mathrm{r}) = \frac{1}{4\pi}\left(\frac{3\mathrm{m}\cdot\mathrm{r}\ \mathrm{r}}{r^5} - \frac{\mathrm{m}}{r^3}\right) , \tag{6}$$

where r denotes a coordinate system centered at the particle. MDM accounts for this change in the magnetic field when calculating the dipole moments. For the ith particle, the moment m_i is then given by [3]:

$$\mathrm{m}_i = C_i\left(\mathrm{H}_0 + \sum_{j=0, j\neq i}^{N} \mathrm{H}_{dip,j}(\mathrm{r}_{i,j})\right) , \tag{7}$$

where $\mathrm{r}_{i,j}$ is the separation distance vector from i^{th} particle to j^{th} particle. Equations 6 and 7 result in a $3N \times 3N$ linear system of equations. The moments of all the particles are coupled and must be solved simultaneously. We discuss numerical methods to solve this system in Sect. 3.2. The calculated moment is then used to calculate the dipolar force (F_i^{dip}) on each particle [3],

$$\mathrm{F}_i^{dip} = -\mu_0 \nabla\left(\mathrm{m}_i \cdot \sum_{j=0, j\neq i}^{N} \mathrm{H}_{dip,j}(\mathrm{r}_{i,j})\right) . \tag{8}$$

2.2 Inclusion model

The Inclusion model provided by Keaveny and Maxey [3] removes the assumption of considering only dipoles. When particles are closer together, other multipoles, in addition to the dipole, become significant. The Inclusion model improves MDM by calculating the SHA solution, which accounts for 'L' multipoles, for particle pairs with separation distances less than a cutoff distance. Below, we give a brief description of the Inclusion model.

Step 1: Calculate forces using a dipolar approach like MDM.

Step 2: Identify the particle pairs with separation distances lower than the cut-off distance.

Step 3: Calculate force using the SHA on the selected particle pairs. There are some important considerations when applying SHA here. The MDM already included the dipolar terms for the selected particle pair. Therefore, Keaveny and Maxey [3] provides some corrective measures to ensure that no calculation is performed twice. Along with dipolar corrections, a component of the magnetic field due to the out-of-plane component of the dipole moment from the MDM needs to be added to the Maxwell Stress Tensor [48]:

$$\mathrm{T} = \mu_0\left(\mathrm{HH} - \frac{1}{2}H^2\mathrm{I}\right) , \tag{9}$$

$$\mathrm{F}_i = \int_{V_i} \nabla\cdot\mathrm{T}\ dV \ . \tag{10}$$

With the corrective terms calculated, the magnetic field is now known. We can use the Eqs. 9 and 10 to calculate the force.

3 Magnetic model implementation

We use LIGGGHTS, an extensively validated SSDEM solver, for our work. In addition to being open-source, multiple force models are available for contact, rolling friction, and cohesion. Various researchers have extended LIGGGHTS with different models, showcasing its ease of adding new models. Finally, the software can scale with parallel processing on High-Performance Computing servers. Therefore, we could focus on implementing the magnetic force model. To ensure correctness, we validated the force models in LIGGGHTS by running unit test cases [50]. The tangential history model needed corrections [36] as a particle didn't stop at the theoretical distance after sliding on a wall.

LIGGGHTS is based on the C++ language and some legacy Fortran-based codes. Users only need to add the corresponding source and header files with the same class style to add a new capability. Various class styles in LIGGGHTS provide a base structure to extend the functionality. For example, the `atom` class deals with all particle-centric

properties like position, velocity, force, and such. For all computations, the source code uses simple C-style data structures. We instead build LIGGGHTS with `eigen` and `boost` libraries to access efficient linear algebra and numerical integration functions.

We defined a new `fix` for the magnetic force model in LIGGGHTS called `fix magnetic`. A fix, in LIGGGHTS, is a modular component applied during time integration to modify particle behavior [51]. It takes six arguments along with the `fix ID` and `group-ID`. The first three arguments define the components of the applied magnetic field H_0 in A/m units in the simulation coordinate frame. Next, we define the magnetic time step (mag_steps), followed by model type (could be `mdm` or `inclusion`) and finally the dipole moment calculation method (could be `convergence` or `linalg`). An example magnetic fix is as follows:

```
fix mag all magnetic 0 0 1 100 mdm convergence
```

The fix currently only works with SI units. It also needs the `atom_style` to be `hybrid granular dipole` as the particles need to be granular and dipolar. Per particle type, the property called `magneticSusceptibility` needs to be defined for this magnetic fix. To run the magnetic model at a different time step (dt_{mag}), we modified the LIGGGHTS `atom` class to add another variable called `mag_f` similar to the existing total force variable `f`. This variable stores the magnetic force for each particle each time a new force is calculated. At all other time steps, the same force is propagated. All the modified `atom` classes and magnetic code (folders src/DIPOLE and src/MAGNETIC) can be found at our public LIGGGHTS fork [52].

3.1 Critical magnetic time

The computation time for the magnetic model is comparatively the highest among the different force models. With the time step being on the order of microseconds ($dt \rightarrow \mathcal{O}(10^{-6})$), the magnetic model can be run at a larger time step. The particles don't move much in a single time step, resulting in minimal change to the magnetic force. This variable time step can help reduce the computational load of the magnetic model. There are different formulations for critical time steps for the Hertzian force model. LIGGGHTS uses the Hertzian collision time (τ_{hc}), which is based on the maximum possible relative velocity of a particle pair v_{max},

$$v_{max} = 2 \max_{i}^{N} |\mathrm{v}_i|, \tag{11}$$

where v_i is the the velocity of ith particle.

$$\tau_{hc} = \min_{i,j}^{N} \left[2.87 * \left(\frac{m_{eff}^2}{a_{eff} * E_{eff}^2 * v_{max}} \right)^{0.2} \right], \tag{12}$$

where effective mass is $m_{eff} = \frac{m_i m_j}{m_i + m_j}$, effective radius $a_{eff} = \frac{a_i a_j}{a_i + a_j}$ and effective Young's Modulus is given using poisons ratio ν

$$\frac{1}{E_{eff}} = \frac{1 - \nu_i^2}{E_i} + \frac{1 - \nu_j^2}{E_j}. \tag{13}$$

Using a similar formulation, we defined a critical time step for the Magnetic force model. Sikka et al. [7] developed an empirical model for the force between two identical sphere paramagnetic particles. They provide three simplified formulas for calculating the magnetic force between particles in specific orientations: parallel to the magnetic field, perpendicular to the field, and touching while parallel to the field. The formula (Eq. 14) for particles parallel to the field provides the maximum possible force between a particle pair. We use this empirical model in our definition of critical magnetic time

$$F = F_{DM} + \mu_0 a^2 H_0^2 \sum_{i=1}^{3} \frac{p_i(\chi)}{c^{4+i}}, \tag{14}$$

$$p_i(\chi) = \sum_{j=0}^{4} q_j^i \chi_{eff}^j \quad i = 1, 2, 3. \tag{15}$$

where $c = r/a$ and the coefficients q_j^i are given in [7]. Given this force, we can find $\frac{dF}{dr}$. We know that the values of F and $\frac{dF}{dr}$ are maximum at $r = 2$ for a given susceptibility. First, we find the change in relative position Δr for which the magnetic force changes by 100%,

$$\Delta r(\chi) \approx \left[\frac{1}{F(\chi)} \frac{dF}{dr}(\chi) \right]^{-1}. \tag{16}$$

Now, we can use the maximum relative velocity to define the critical magnetic time (τ_{mag}),

$$v_{max} \tau_{mag} \approx \left[\frac{1}{F(\chi)} \frac{dF}{dr}(\chi) \right]^{-1}. \tag{17}$$

We have implemented this into our LIGGGHTS model using a `fix check/timestep/mag` command. The user inputs the tolerable $\frac{\Delta F}{F}$ and the time step at which they

want to check the magnetic time. It is worth noting that the empirical formula is based on a spherical harmonic approximation. Therefore, the critical magnetic time τ_{mag} would be much smaller than the exact definition using the FDM or MDM model. A larger tolerance is advised when using the MDM model. We compare the critical magnetic time with the Hertz collision time in one set of our simulations as discussed in Sect. 5.4.

3.2 MDM implementation

The MDM method described in Sect. 2.1 requires a solution to $3N \times 3N$ linear system. For large systems with $\approx 1000s$ of particles, even optimized C++ libraries like `eigen` require a lot of computational resources. One possible solution is to limit the dipole influence of each particle (eq. 6) as it directly scales with the cube of the distance from the particle (r^3). Fazekas et al. [10] employ a similar technique for force calculation using the FDM by keeping the magnetic force cutoff at $\approx 6.25D$ where D is the particle diameter. Han et al. [4] show that for an infinite chain of particles, a force cutoff of $\approx 6a$ gives less than 5 % relative error. While force cutoff is a possible solution for the FDM model and dispersed/special systems, we observed that the aggregate of all small interactions from far-field particles becomes significant for a large particle system. Leps and Hartzell [6] implemented an MDM solver in LIGGGHTS while using a cut-off distance. They used a convergence algorithm instead of solving the linear system.

However, their convergence algorithm is not guaranteed to converge each time. Han et al. [4] proposed a convergence algorithm using the Gauss-Seidel method. This method uses the latest dipole moment value rather than the value from the last step as described in the following equation.

$$\mathrm{m}_i^{k+1} = \mathrm{m}_i^0 + C_i \sum_{j=0, j\neq i}^{N} \frac{(3\hat{\mathbf{r}}_{ij}.\hat{\mathbf{r}}_{ij}^T - \mathbb{I})}{4\pi r_{ij}^3} \mathrm{p} \ , \tag{18}$$

$$\text{where } \mathrm{p} = \begin{cases} \mathrm{m}_j^{k+1} & \text{if } j < i \\ \mathrm{m}_j^k & \text{if } j > i \ . \end{cases} \tag{19}$$

By stacking the dipole moments of each particle, we can create a stacked moment vector $\mathfrak{m} = [\mathrm{m}_i]^T$, and we can define a new $3N \times 3N$ matrix called $\mathbb{R}$ whose 3×3 elements at $3i$, $3j$ can be defined as $\mathbb{R}_{3i,3j} = C_i \frac{(3\hat{\mathbf{r}}_{ij}.\hat{\mathbf{r}}_{ij}^T - \mathbb{I})}{4\pi r_{ij}^3}$ to result in the following linear discrete-time system,

$$\mathfrak{m}_{k+1} = \mathfrak{m}_0 + \mathbb{L}\ \mathfrak{m}_{k+1} + \mathbb{U}\ \mathfrak{m}_k \ , \tag{20}$$

where $\mathbb{L}$ and $\mathbb{U}$ are lower and upper triangular matrices of $\mathbb{R}$ respectively. As $\mathbb{R}$ has a zero diagonal, $\mathbb{L}$ and $\mathbb{U}$ are strictly triangular matrices. By applying the same technique to this system, we need to examine the eigenvalues of $(\mathbb{I} - \mathbb{L})^{-1}\mathbb{U}$. Using that $\mathbb{L}$ and $\mathbb{U}$ are strictly triangular with all eigenvalues equal to zero. It can be shown that $(\mathbb{I} - \mathbb{L})^{-1}\mathbb{U}$ has all eigenvalues equal to 0. Therefore, the system converges.

In our implementation (Algorithm 1), we use this discrete-time system given in Eq. 18. Different columns of the $\mathbb{R}$ matrix are calculated by individual processors as per their local particles. Then, the convergence algorithm is applied on processor 0, after which the converged moments are distributed to all the processors.

Algorithm 1 Convergence-based Mutual DipoleMethod for LIGGGHTS

initialization;
each local processor calculates column matrices for $\mathbb{R}$ *matrix*
for every i^{th} local particle **do**
  for $j = 0$ **to** $N - 1$ **do**
    if $i \neq j$ **then**
      calculate the 3×3 matrix $\mathbb{R}_{3j,3i}$;
    end if
  end for
  stack $\mathbb{R}_{3j,3i}$ for $j = 0$
end for
Gather all local matrices to create the $\mathbb{R}$ matrix on proc 0;
if `proc` $== 0$ **then**
  k_{max} = max number of convergence steps;
  ϵ = convergence threshold;
  for $k =$ **to** $k_{max} - 1$ **do**
    for $i =$ **to** $N - 1$ **do**
      $\mathbf{m}_i^0 = C_i \mathbf{H}_0$;
      $\mathbf{m}_i^k = \mathbf{m}_i^0$;
      for $j = 0$ **to** $N - 1$ **do**
        if $i < j$ **then**
          $\mathbf{m}_i^k = \mathbf{m}_i^k + \mathbb{R}_{i,j}\mathbf{m}_j^{k-1}$;
        else if $i > j$ **then**
          $\mathbf{m}_i^k = \mathbf{m}_i^k + \mathbb{R}_{i,j}\mathbf{m}_j^k$;
        end if
      end for
      $m_{dif}(i) = \frac{||\mathbf{m}_i^k - \mathbf{m}_i^{k-1}||}{C_i||\mathbf{H}_0||}$;
    end for
    if $\max_i m_{dif}(i) < \epsilon$ **then**
      break ;
    end if
  end for
end if

We also implement another algorithm to solve the $3N$ linear systems of equations described in equations 6 and 7. It follows the same principle as the `convergence` algorithm. The $3N \times 3N$ linear system can be written in terms of the earlier defined $\mathbb{R}$ matrix by modifying the discrete-time system,

$$\mathfrak{m} = \mathfrak{m}_0 + \mathbb{R}\ \mathfrak{m}\ , \tag{21}$$

$$(\mathbb{I} - \mathbb{R})\mathfrak{m} = \mathfrak{m}_0\ . \tag{22}$$

Each processor calculates the corresponding components (for local particles) of both $(\mathbb{I} - \mathbb{R})$ and $\mathfrak{m}_0$. These are gathered on processor 0, and the linear system is solved using the `eigen` library's `ColPivHouseholderQR` algorithm, which is based on the Householder rank-revealing QR decomposition of a matrix with column pivoting. The calculated moments are then scattered to the respective processors for housekeeping.

LIGGGHTS uses neighbor lists to track particle pairs and potential interactions between them. These lists are generated by each processor for its local particles. Therefore, every processor only stores information for local particles and their neighbors. While this setup is both memory and computationally efficient for other DEM force models, our magnetic model requires all particle pairs to be considered. We modify the neighbor parameters for our simulations to generate neighbor lists that include all particle pairs.

To ensure all particle pairs are considered, we need the neighbor cutoff distance, known as `neighbor skin` distance in LIGGGHTS, to encompass the whole simulation domain. We refer to a neighbor list with all particle pairs as a complete neighbor list. This larger `skin` distance has some drawbacks. Each processor has its own neighbor list for each local particle. If neighbors for a particle are local to another processor, information pertaining to those particles is shared between the processors. With a complete neighbor list, there is a significant communication overhead. Therefore, it is essential to optimize the number of processors used.

Another critical parameter is the frequency of neighbor list generation. For a complete neighbor list, we could, in theory, never update the neighbor list after the simulation setup. However, LIGGGHTS also decides if a particle should be moved between processors during neighbor list generation. If the neighbor list is not updated, all particles remain associated with their initial processors, even if they have moved significantly within the simulation domain. At the end of the simulation run, LIGGGHTS identifies the particles that have moved significantly, determines they are behaving abnormally, and deletes them. Any restart file generated after the run would not contain the said particles. Therefore, the frequency of neighbor list generation also needs to be optimized.

3.3 Force calculation

Once the moments are calculated with either `linalg` or `convergence` (Algorithm 1), we can calculate the dipolar force between particles. The force between two dipolar particles can be calculated as [12]:

$$\mathrm{F}_{i,j}^{dip} = \frac{3\mu_0}{4\pi r^4}\Big[(\mathrm{m}_i.\hat{\mathrm{r}})\mathrm{m}_j + (\mathrm{m}_j.\hat{\mathrm{r}})\mathrm{m}_1\ , + (\mathrm{m}_i.\mathrm{m}_j)\hat{\mathrm{r}} - 5(\mathrm{m}_i.\hat{\mathrm{r}})(\mathrm{m}_j.\hat{\mathrm{r}})\hat{\mathrm{r}})\Big] \tag{23}$$

where $\mathrm{r} = \mathrm{r}_j - \mathrm{r}_i$, $r = |\mathrm{r}|$ and $\hat{\mathrm{r}} = \mathrm{r}/r$.

Therefore, the force on a single particle is the sum of the forces on all other particles,

$$\mathrm{F}_i^{dip} = \sum_{j=0, j\neq i}^{N-1} \mathrm{F}_{i,j}^{dip}\ . \tag{24}$$

The force due to the spherical harmonics approximation (F_i^{sha}) for the Inclusion model (discussed more in the next section) is calculated for particle pairs with separation distances less than four radii ($r_{i,j}/a_i = c < 4$). Therefore, the total magnetic force on the ith particle (Algorithm 2) is calculated,

$$\mathrm{F}_i^{mag} = \mathrm{F}_i^{dip} + \mathrm{F}_i^{sha}\ . \tag{25}$$

The calculated force F_i^{mag} is then assigned to `atom` class variables for total force (`f`) and magnetic force (`mag_f`).

Algorithm 2 Force Calculation

initialization;
if dt **modulo** mag_steps **then**
  for every i^{th} local particle **do**
    `f[i] += mag_f[i];`
  end for
else
  for every i^{th} local particle **do**
    for $j = 0$ **to** $N - 1$ **do**
      if $i \neq j$ **then**
        calculate $\mathbf{F}_{i,j}^{dip}$
        $\mathbf{F}_i^{dip} = \mathbf{F}_i^{dip} + \mathbf{F}_{i,j}^{dip}$
        if $r_{i,j} < 4a_i$ **and** $model_type ==$ `'inclusion'` **then**
          calculate $\mathbf{F}_{i,j}^{sha}$
          $\mathbf{F}_i^{sha} = \mathbf{F}_i^{sha} + \mathbf{F}_{i,j}^{sha}$
        end if
      end if
    end for
    $\mathbf{F}_i^{mag} = \mathbf{F}_i^{dip} + \mathbf{F}_i^{sha}$
    $\mathtt{mag_f[i]} = \mathbf{F}_i^{mag}$
    $\mathtt{f[i]} += \mathtt{mag_f[i]}$
  end for
end if

3.4 Inclusion model implementation

As described in the force calculation algorithm 2, the `model_type inclusion` only affects the calculation of $\mathrm{F}_{i,j}^{sha}$. We use a cutoff distance to ascertain when to incorporate multipoles for a particle pair by calculating the SHA force. Sikka et al. [7] and Fig. 5 show that MDM and SHA converge quickly for a two-particle case. Therefore, we finalized a cutoff distance of three radii ($3a$). We have a separate class called `spherical_harmonics` to calculate the SHA force. The initializer takes six input arguments: the particles' radius (a), the particles' susceptibility (χ), the applied magnetic field vector (H_0), the separation distance vector ($\mathrm{r}_{i,j}$) and finally the respective dipole moments (m_i and m_j).

The `spherical_harmonics` class initializes the two $2L \times 2L$ linear systems needed for spherical harmonics approximation. Table 1 is used to find the number of multipoles L to be used for a given susceptibility χ. The corrections to the system are added to account for the dipolar force that has already been calculated.

In a two-body isolated case, each particle would only have in-plane components (plane with the two particles and the external magnetic field) of the dipole moments. We therefore need to ensure that the corresponding spherical harmonics (β_{11} and β_{10} in equation A2) don't have the dipolar force calculated twice. The bulk scenario, instead, would have an out-of-plane component of dipole moment. Keaveny and Maxey [3] include a new term in the Inclusion model based on this out-of-plane component. One can think of this as adding a β_{12} into the Eq. A2. But, there are no such terms in any other multipoles in the SHA formulation; therefore, it results in non-physical phenomena in bulk simulations. This phenomenon is further discussed in detail in the sub-sect. 3.4.1. The validation cases discussed in [3] are all particle chains with all out-of-plane components equal to zero. We remove this term in our implementation.

The linear system is then solved using the `ColPivHouseholderQR` algorithm mentioned earlier. The surface integral of the Maxwell stress tensor (Eq. 10) was first reduced analytically using Mathematica. The reduced polynomial dependence is then solved with `boost` library's Gaussian Konrod algorithm. The Spherical Harmonics Approximation and the reduced polynomials are discussed in detail in the Appendix A.

3.4.1 Out-of-plane dipolar correction

The out-of-plane dipole correction is applied to the SHA force to add additional information from the out-of-plane component of the dipole moment calculated during the MDM method. The scalar potential $\Phi_{out-plane}$ is given by [3]:

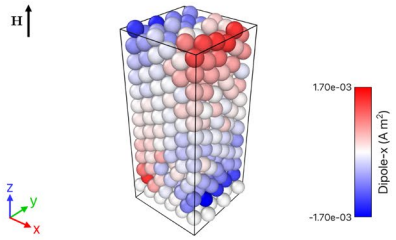
$$\Phi_{out-plane,i} = \frac{\mathrm{m}_i.\hat{\mathrm{y}}}{4\pi a^3} \frac{P_l^m \cos\theta_i \sin\phi_i}{r_i^2} , \tag{26}$$

where P_l^m is the associated Legendre polynomial, m_i is the dipole moment and r_i, θ_i and ϕ_i are the spherical coordinates of the ith particle respectively. The correction to the magnetic field is then the sum of the gradient of scalar potentials of the two particles, i and j:

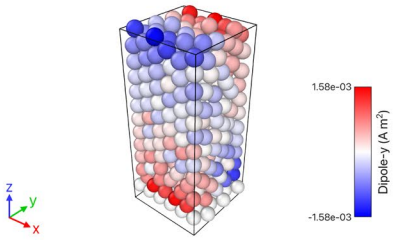
$$\mathrm{H} = -\nabla\Phi_{out-plane,i} - \nabla\Phi_{out-plane,j} . \tag{27}$$

Table 1 Number of Multipoles (L) for a given magnetic susceptibility (χ). These values were obtained by testing the two-particle configuration (Sect. 4.1) for different susceptibilities and finding the minimum number of multipoles required for convergence

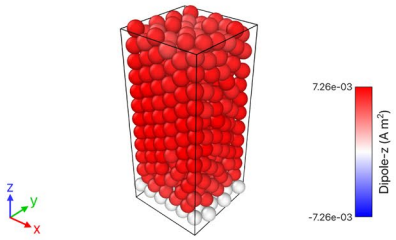
χ	(0, 4)	[4, 7)	[7, 12)	[12, 20)	[20, 35)	$[35-55)$	[55, 75)	$[75-100)$
L	10	20	30	40	45	50	75	80



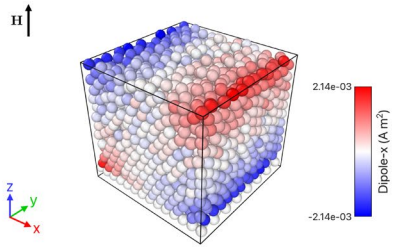
(a) Size 1: x-component of dipolar moment



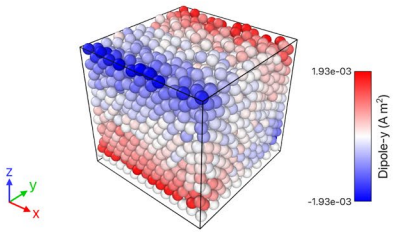
(b) Size 1: y-component of dipolar moment



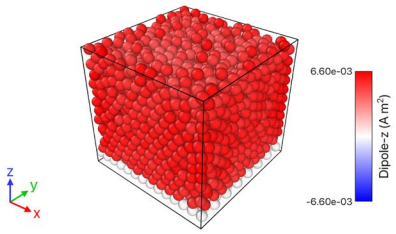
(c) Size 1: z-component of dipolar moment



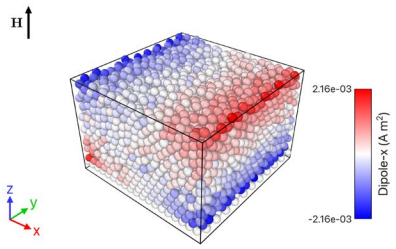
(d) Size 2: x-component of dipolar moment



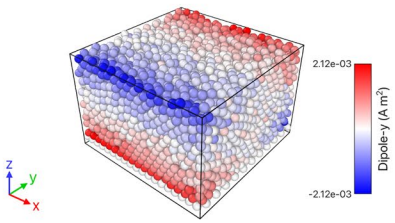
(e) Size 2: y-component of dipolar moment



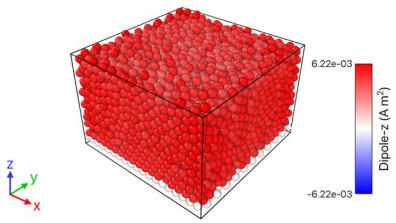
(f) Size 2: z-component of dipolar moment



(g) Size 3: x-component of dipolar moment



(h) Size 3: y-component of dipolar moment



(i) Size 3: z-component of dipolar moment

Fig. 2 Dipolar moments of particles for three different sizes: Size 1: $35 \times 35 \times 130$ mm, Size 2: $85 \times 85 \times 130$ mm, and Size 3: $120 \times 120 \times 130$ mm. The applied magnetic field is in the positive z-direction (upwards). The x-component of the dipole moment is shown to be opposite near opposing x-walls (2a, 2d, 2g), while the y-component is opposite near opposing y-walls (2b, 2e, 2h). These figures illustrate the effect of the out-of-plane dipolar moment term in the Inclusion model, leading to spiral motion. Therefore, we require the modification of removing the out-of-plane correction term as described in Sect. 3.4.1

We can use simple geometric relations to convert the spherical coordinates of one particle to another by including the separation vector c. We observed a unique non-physical phenomenon once we implemented these equations into our model. The top particles are further from the center of the box in a spiral motion. To understand this phenomenon, we simulated boxes of different sizes. We used three different box sizes: Size 1: $35 \times 35 \times 130$ mm, Size 2: $85 \times 85 \times 130$ mm, and Size 3: $120 \times 120 \times 130$ mm with the same particle size (6.35 mm diameter) and susceptibility ($\chi = 3.6$). The magnetic field is applied in the $+z$ direction, with all other parameters held constant across the three boxes.

Figure 2 shows the dipolar moments of the particles. We see that the x-component of dipole moment (Fig. 2a, 2d, 2g) have opposite signs near opposing x-walls while the y-component of dipole moment (Fig. 2b, 2e, 2h) have opposite signs near opposing y-walls. The z-component of the dipole moment is aligned in the positive z-direction, the same as the applied magnetic field. These dipole moments contribute to the out-of-plane term, leading to a spiral motion.

In Fig. 3, we plot the specific angular momentum about the center of the box of top particles against their distance from the center for each of the three boxes. Most particles have positive angular momentum, indicating rotational

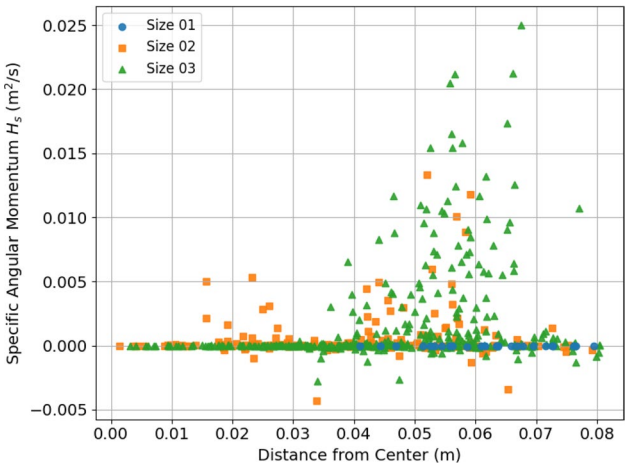
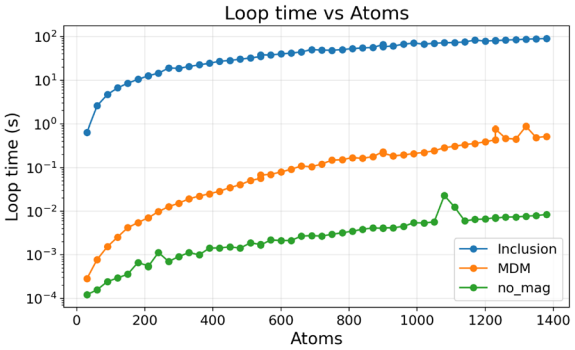


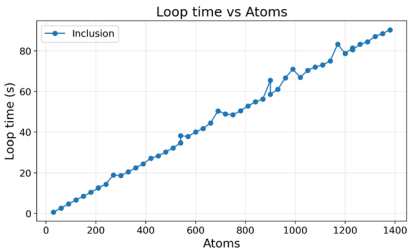
Fig. 3 Specific Angular Momentum about the center of the box of top particles against their distance from the center for three different box sizes at a representative time step. The H_s increases as the distance of the particles from the center increases, indicating the spiral motion around the center of the box. Size 01: $35 \times 35 \times 130$ mm, Size 2: $85 \times 85 \times 130$ mm and Size 3: $120 \times 120 \times 130$ mm

motion about the center of the box. For the smallest box, the top particles quickly settle out of the spiral motion because the walls are too close for the given particle radius. We can observe that the particles at the center (further away from the wall) have almost zero velocities for other sizes. In contrast, the particles closer to the walls have higher velocities. Particles closest to the walls have lower angular momentum because they are slowed down by friction with the walls. Therefore, we observe a peak in angular momentum at some distance between the center and the wall.

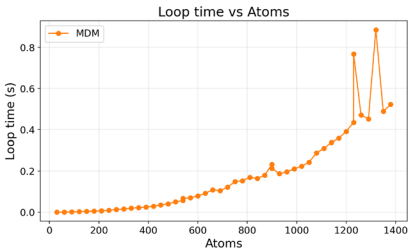
We conclude that including the out-of-plane term in the SHA implementation leads to this nonphysical rotation because the MDM model lacks the corresponding out-of-plane terms. This could be achieved by including another multipole in the far-field calculation and the MDM. We remove this out-of-plane correction term, as the aim of this study is to implement a more accurate model than dipolar models such as MDM.



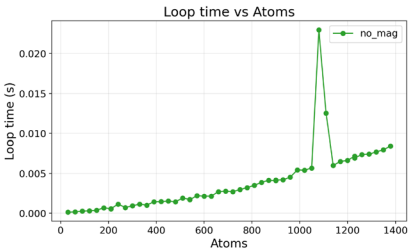
(a) Logarithmic scale of loop time for all three cases.



(b) Loop time for Inclusion model.



(c) Loop time for MDM model.



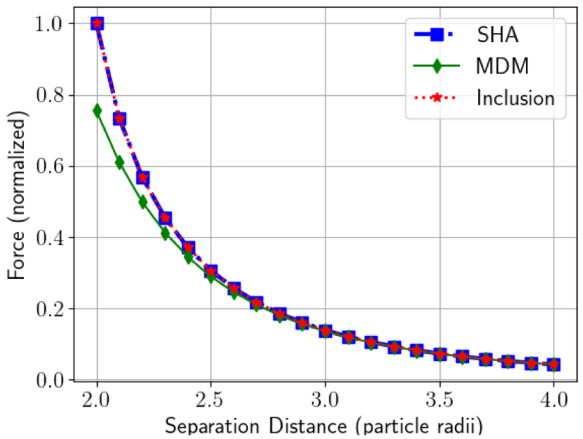
(d) Loop time for no magnetic force case.

Fig. 4 Computational effort of the MDM and Inclusion model for different numbers of particles in an HCP configuration. The computational effort of no magnetic force calculation is also included for reference. The MDM model is more computationally efficient than the Inclusion model and scales quadratically with the number of particles, while the Inclusion model scales linearly. The loop times for the Inclusion model are approximately two orders of magnitude higher than the MDM model, while the MDM model is approximately one order of magnitude higher than the no magnetic force case

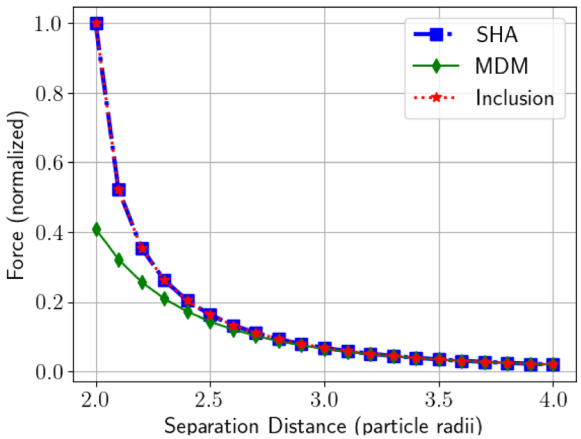
3.5 Computational comparison

We compared the computation cost of the MDM (convergence-based) and the Inclusion model by systematically increasing the number of particles in hexagonal close-packed (HCP) configurations. A box of dimensions $10 \times 10 \times 100$ m was used for this comparison study. The particles were arranged in an HCP configuration with a particle radius of 1 m and a susceptibility of $\chi = 1$. The magnetic field was applied in the z-direction, and the force was calculated twice for each configuration and each model. We increased the number of particles by increasing the number of layers in the z-direction while keeping the x and y dimensions constant. There is no particle motion in this computational comparison. The results for loop times (total run-time in LIGGGHTS output file) are shown in the Fig. 4. We also include the computational effort of the no-magnetic force calculation for reference.

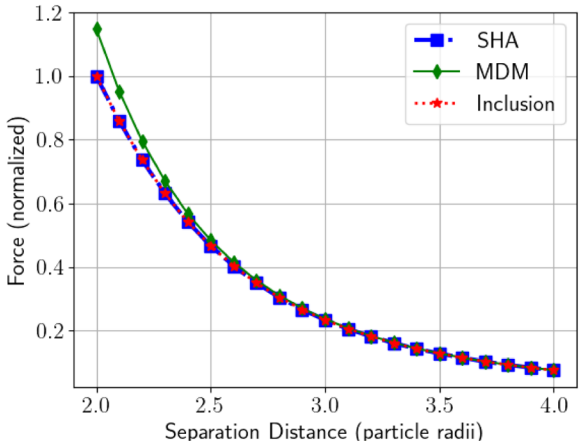
It is clear that the MDM model is much more computationally efficient than the Inclusion model. The loop times for the Inclusion model are approximately two orders of magnitude higher than the MDM model, while the MDM model is approximately one order of magnitude higher than the no magnetic force case. The computational effort for the Inclusion model is linear, while the MDM model is quadratic with the number of particles, as expected. These results are for a specific configuration and may vary for different configurations, but they provide a general idea of the computational effort required for each model. The MDM model, being more computationally efficient, is suitable for larger simulations with sparse particle configurations as the accuracy of the MDM model is comparable to the Inclusion model for such cases, as shown in the validation cases in the



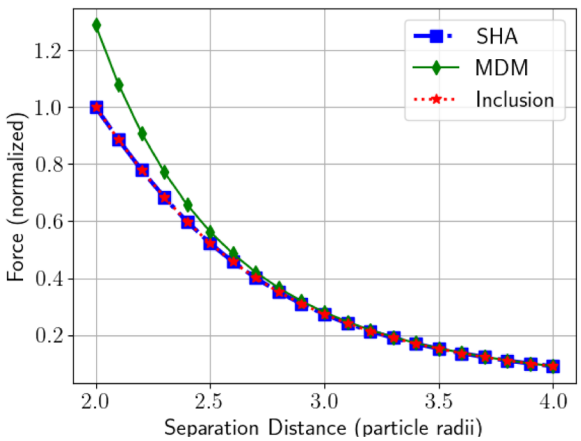
(a) $a = 1$ m, $H_0 = 1$ A/m, $\alpha = 00^{\circ}$, $\chi = 1.0$



(b) a=1 m, H_0=1 A/m, $\alpha = 00^{\circ}$, $\chi = 4.0$

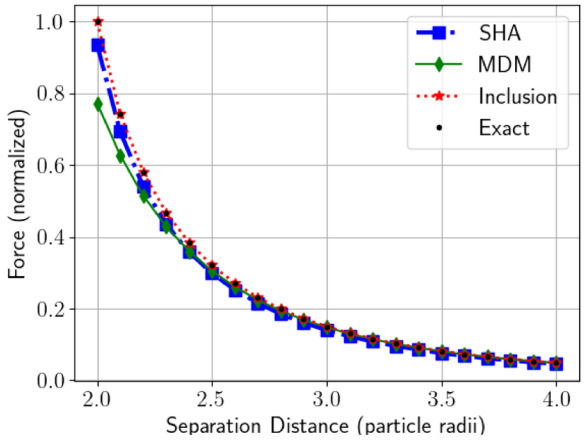


(c) a=1 m, H_0=1 A/m, $\alpha = 90^{\circ}$, $\chi = 1.0$

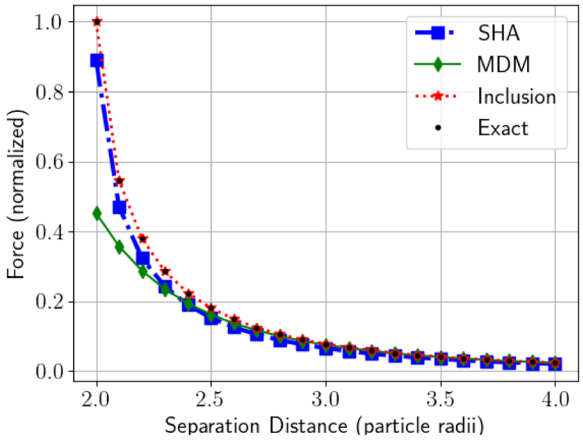


(d) a=1 m, H_0=1 A/m, $\alpha = 90^{\circ}$, $\chi = 4.0$

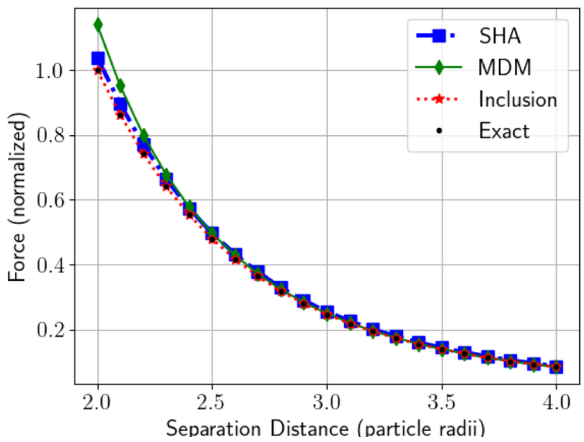
Fig. 5 Validation curves for a two-particle configuration. The force is normalized by the value when the particles are touching. For higher susceptibility, the MDM underestimates the force when particles have a separation distance less than 2.5 particle radii. SHA is considered the exact force in this two-particle case



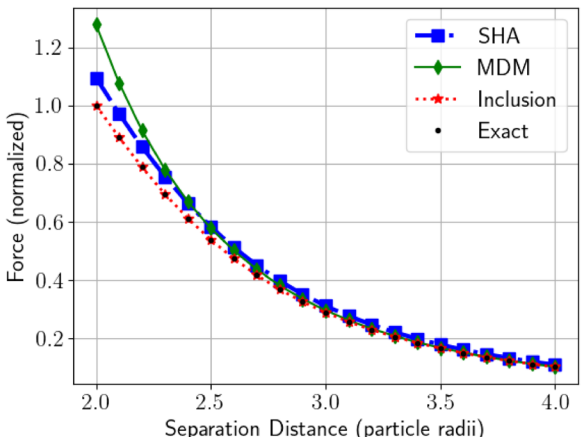
(a) a=1 m, H_0=1 A/m, $\alpha = 00^{\circ}$, $\chi = 1.0$



(b) a=1 m, H_0=1 A/m, $\alpha = 00^{\circ}$, $\chi = 4.0$



(c) a=1 m, H_0=1 A/m, $\alpha = 90^{\circ}$, $\chi = 1.0$



(d) a=1 m, H_0=1 A/m, $\alpha = 90^{\circ}$, $\chi = 4.0$

Fig. 6 Validation curves for a three-particle configuration. The force is normalized by the value when the particles are touching. For higher susceptibility, the MDM underestimates the force when particles are closer than 2.5 particle radii, while the SHA is closer to the exact solution, and the Inclusion model matches the exact force closely

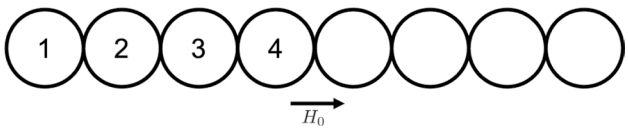


Fig. 7 Eight-particle chain; applied with aligned with axis of the chain ($\chi = 1;\ H_0 = 1$)

next section. For dense particle configurations, the Inclusion model is more accurate, but the computational effort may be prohibitive for larger simulations.

4 1-D validations

We validated the model against simple 1-D test cases for which the results are available in the literature [3, 7, 12]. We tested both algorithms (`convergence` & `linalg`) for the MDM model. Results for these always matched; therefore, for the rest of the paper, MDM refers to the convergence-based algorithm only.

4.1 Two-particle case

We first tested the magnetic force model for two particles in two configurations by varying the inter-particle distance to compare it with [3] and [7]. We also show (Fig. 5) the results for the SHA model. The angle α is varied to test different orientations of the magnetic field with respect to the line joining the two particles. We tested two orientations: parallel ($\alpha = 0°$) and perpendicular ($\alpha = 90°$). SHA is considered to be an exact result for the two-particle case when a large number of multipoles are considered ($L \approx 20$ & 40 for $\chi = 1$ & 4 respectively) [3]. The Inclusion model matches exactly with SHA, as no other far-field force exists. The

Table 2 MDM Results for 8 particle chain configuration: Expected force (N) [3] and percentage error (against expected force) for the various particles (Fig. 7) for different average neighbors. The error percentage is calculated as $\frac{F_{expected}-F_{calculated}}{F_{expected}} \times 100\%$

Particle	Expected	Average number of neighbors per particle				
Number	Force (N)	4.5	5.5	6.25	6.75	7
1	5.0252E-07	−1.04%	−0.45%	−0.19%	−0.06%	0.00%
2	7.5187E-08	−4.09%	−1.30%	−0.21%	0.11%	0.00%
3	1.8014E-08	−12.60%	−1.82%	0.90%	0.06%	0.00%
4	3.9195E-09	−39.78%	6.44%	0.46%	0.06%	0.01%

Table 3 Inclusion Results for 8 particle chain configuration: Expected force (N) [3] and percentage error (against expected force) for the various particles (Fig. 7) for different average neighbors. The error percentage is calculated as $\frac{F_{expected}-F_{calculated}}{F_{expected}} \times 100\%$

Particle	True	Expected	Average number of neighbors per particle				
Number	Force (N)	Force (N)	4.5	5.5	6.25	6.75	7
1	6.38E-07	6.4659E-07	−1.01%	−0.53%	−0.31%	−0.21%	−0.16%
2	7.07E-08	7.9784E-08	−3.84%	−1.20%	−0.18%	0.10%	−0.02%
3	1.72E-08	1.8894E-08	−11.75%	−1.48%	0.93%	0.03%	−0.03%
4	3.84E-09	4.0966E-09	−37.25%	6.88%	0.43%	0.02%	−0.02%

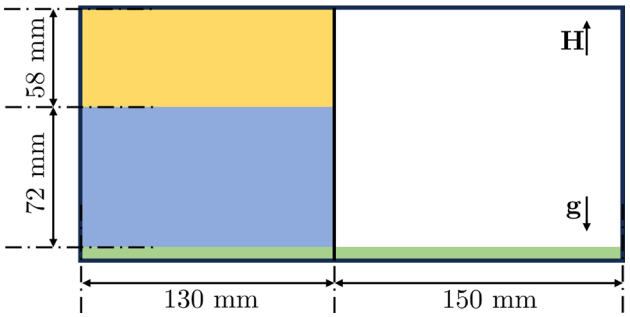


Fig. 8 Mass wasting setup: Green section are the fixed floor particles in a hexagonal packing, blue section is filled with the magnetic particles for avalanching and the yellow section is used to insert particles while preparing the particle assembly

MDM underestimates the force when particles are closer together in the case of parallel magnetic field orientation, while it overestimates the force for perpendicular orientation. We also tested the quadratic dependence [7] on particle radius (a) and applied magnetic field magnitude (H_0) as other test cases consider unit values for these parameters.

4.2 Three-particle case

For the three-particle configuration, we fix two particles to be touching, and the third particle's distance is varied along the line connecting all three particles. The force on the moving particle is plotted. The results of this test case, shown in Fig. 6, match with [3] where they suggest a pair-wise Spherical Harmonic Approximation with more multipoles ($L = 20$ for $\chi = 1$ and $L = 30$ for $\chi = 4$) is equivalent to the exact force. The Inclusion model closely matches the exact force for all four cases considered. The MDM model is very inaccurate for higher susceptibilities ($\chi = 4$) in both external magnetic field orientations (parallel and perpendicular to the line joining the particles).

4.3 Eight-particle case

Keaveny and Maxey [3] also tested the Inclusion model for an eight-particle chain depicted in Fig. 7. We reproduce the results with our model here in Tables 2 and 3. We tried different `neighbor skin distances` (Force cut-off in LIGGGHTS), resulting in different average number of neighbors per particle. These tests were our first indication that a force cut-off for magnetic force could result in an inaccurate calculation. Considering a lower number of average neighbors results in higher error percentages. Especially when the cut-off distance was less than 4 particle diameters (average neighbors: 5.5), the error percentages were close to 40% for the middle particles. Both the MDM and Inclusion models exhibited this behavior. When all particles are considered as each other's neighbors (average neighbors: 7.0), we observed that our model matches the expected force value [3]. Therefore, we concluded that the magnetic force model is sensitive to the neighbor skin distance. In our subsequent simulations, we considered all particle neighbor pairs.

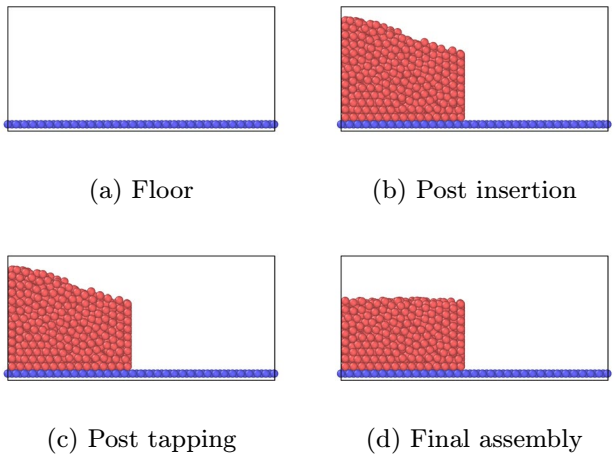


Fig. 9 Different stages of particle assembly preparation to obtain a desired packing fraction

Table 4 Material Properties for Avalanching [8, 54, 55]

Quantity	Symbol	Units	Steel	Floor (Brass)	Wall (Acrylic)
Density	ρ	(kg/m^3)	7800	8500	1250
Young's modulus	E	(MPa)	2100	1100	30
Poisson ratio	ν	(-)	0.3	0.375	0.4
Magnetic Susceptibility	χ	(-)	$3.6-40.0$	0	0

Table 5 Interaction Properties for Avalanching [10, 53]

Quantity	Symbol	Units	Steel-Steel	Steel-Floor	Steel-Wall
Coeff. of restitution	e	(-)	0.80	0.70	0.40
Friction coefficient	μ_s	(-)	0.2	0.15	0.10
Rolling friction coeff	μ_r	(-)	0.02	0.015	0.01
Rolling visc. damping	$\gamma_{d,r}$	(-)	0.02	0.02	0.02
Cohesion energy density	k_c	(mJ/m^3)	2	2	2

5 Avalanching of metallic grains

After validating the magnetic force model with simple 1-D cases, we tested the model against more complex scenarios. Sunday et al. [8] performed avalanching experiments with brass and steel balls placed between two Helmholtz coils to simulate a uniform magnetic field. We modeled the same setup in LIGGGHTS to reproduce the bulk behavior of metallic grains with our model. Previous avalanche studies have examined the effects of magnetic cohesion on the angle of repose of steel grains through experimental [9] and numerical [10, 11] investigations. Sunday et al. [8] performed experiments with different particles and a larger range of magnetic field strength than Forsyth et al. [9]. Fazekas et al. [10] & Taylor et al. [11] used a dipolar algorithm for force calculation, which is inaccurate at closer distances, as shown in Sect. 4. They could not reproduce the increase in angle of repose or the bulk flow regimes, such as the correlated and uncorrelated regimes. Our implementation can be used with either the dipolar algorithm or the Inclusion model. We tested both algorithms to compare their results.

5.1 Setup

Sunday et al. [8] uses 3.18 mm, 6.35 mm, and 8.73 mm-sized grains. We looked at the experiments with 8.73 mm grains, as fewer particles provide a reasonable computational time. Fig. 8 shows the simulation setup. An avalanche is triggered by removing the middle partition wall. The length of the box (x-axis) is 280 mm, with a middle partition at 130 mm. The width (y-axis) is 85 mm while the height (z-axis) is initially set to 130 mm. The floor is modeled with brass particles (Fig. 9a) that are frozen in a hexagonal closed packing to simulate a coarse floor. We use fixed boundaries in LIGGGHTS with granular walls (`fix wall/gran`) to keep particles within the simulation box. The magnetic susceptibility for walls is set to zero to simulate non-magnetic walls. The middle partition wall is also modeled similarly. The gravity is in the downward ($-z$) direction while the applied magnetic field is in the upward ($+z$) direction. Once particle assembly is prepared, we change the ceiling (`change_box`) to a shrink-wrapped boundary condition to have fewer idle processors. The middle partition is removed (`unfix`) to initiate the avalanche.

5.2 Material properties

We used three materials in the simulation: alloy steel for magnetic particles, brass for the floor particles, and acrylic for the walls, including the middle partition. The material properties are given in Table 4. The densities (ρ) for alloy steel and brass are modeled the same as [8]. Multiple DEM studies [36, 53] have shown that using a factor of 0.01−0.0001 to the actual Young's modulus (E_0) does not affect bulk flow behavior. The Hertz collision time, and therefore time-step (dt), however, directly depends on Young's modulus ($dt \propto \sqrt{E}$). We use a 0.01 to E_0 factor for our E values to keep computation time reasonable. The mechanical properties of the walls were based on PMMA material [53].

The interaction properties between different materials, the coefficient of restitution (e), and the rolling friction coefficients (μ_r, $\gamma_{d,r}$) were based on DEM studies by Hlosta et al. [53]. The friction coefficient varied between 0.3 [53] and 0.7 [10]. We start with the values by [53] for all interaction

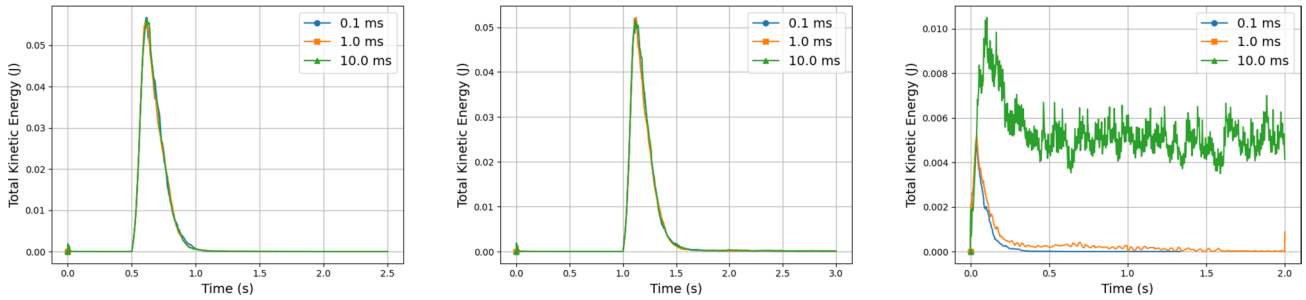


Fig. 10 Sensitivity analysis for magnetic time step (mag_t) with 6.35 mm particles, susceptibility of 40.0, and field strengths of 5.0 mT (Left), 15.0 mT (Middle), and 30.0 mT (Right). The magnetic time step of 10.0 ms never settles for a 30 mT field strength, while the 1.0 ms and 0.1 ms magnetic time steps settle for all three field strengths. The 1.0 ms and 0.1 ms magnetic time steps have similar kinetic energy profiles for all three field strengths

properties. We then calibrate the interaction properties' values to match the angle of repose of experiments [8] in the absence of any magnetic field. The calibrated values are mentioned in Table 5.

We set the magnetic susceptibility for the walls and the brass particles to zero because they are assumed to be non-magnetic. Two types of susceptibility values are mentioned in the literature: bulk susceptibility and material susceptibility. Material susceptibility is intrinsic to the material, while bulk susceptibility depends on the shape and size of the material sample. Sunday et al. [8] mentions the bulk effective susceptibility of the coarse alloy grains to be 3.59 ± 0.12. However, the material susceptibility is mentioned to range between 13 and 100. Sunday et al. [8] observed that the steel particles demonstrated plastic flow behavior starting at 25 mT field strength. We varied the material susceptibility between 3.6 and 40.0 at a field strength of 25 mT. With the Inclusion model, we could reproduce the flow behavior with a material susceptibility of 20.0.

5.3 Particle assembly preparation

Each simulation starts with preparing a particle assembly. We use the same particle size for the floor (brass) and the bulk (steel) particles. The particles are modeled as spheres with a diameter of 8.73 mm (coarse). We arrange the floor particles in a hexagonal packing using LIGGGHTS' `lattice` command (Fig. 9a). The left half is filled by pouring particles and allowing them to settle under gravity, as it provides more control over the packing fraction (φ) [36]. The packing fraction is defined as the ratio of the volume of particles to the total bulk volume. For packing fraction calculation, we only consider the assembly region up to the height of 72 mm (blue in Fig. 8). The particles are inserted using `ins/rate/region` fix in LIGGGHTS. We define an insertion region (yellow in Fig. 8), insertion rate, total number of particles inserted, and a random seed. The particles are inserted at a rate of 2 per time step (dt). A slow insertion rate facilitates a higher packing fraction of the assembly and, therefore, a compact particle assembly. We aim for a packing fraction of $\varphi \approx 0.59$ to match experiments [8]. The extra particles are then deleted to ensure a fixed height (72 mm).

Table 6 Critical magnetic time step (τ_{mag}) and hertz collision time (τ_{hc}) for different field strengths at $0.1ms$ magnetic time step (mag_t) for our time sensitivity analysis. The ratio of the two is also shown

H (mT)	τ_{hc} (s)	τ_{mag} (s)	Ratio ($\frac{\tau_{mag}}{\tau_{hc}}$)
5.0	2.176272e−05	1.121155e−01	5.151722e+03
15.0	2.215137e−05	1.224906e−01	5.529708e+03
30.0	2.414839e−05	1.885994e−01	7.810023e+03

Table 7 Time steps used in the simulations

Quantity	Symbol	Units	Value
Time step	dt	μs	5
Magnetic time step	mag_t	ms	1
Neighbor time step	(-)	ms	10

In some cases, we tap the assembly to achieve the required packing fraction (Fig. 9c). The assembly floor is provided with a sinusoidal vibration: $z_t = z_0 + A\sin(2\pi ft)$ [36]. We use the same amplitude ($A = 0.5$ mm) and frequency ($f = 30$ Hz) as used by Frizzell and Hartzell [36]. The tapping is done for fixed intervals, and the packing fraction is calculated. Once the desired packing fraction is achieved, we stop the tapping. The final particle assembly (Fig. 9d) is then used to perform various avalanching simulations using a restart file (`write_restart` command). We provide input scripts and restart files in our dataset [56]. Multiple seeds are used to generate different assemblies for the same packing fraction.

For each simulation, we first allow the particles to settle after switching on the external magnetic field. By 'settle', we mean particles should reach a steady state. We define the steady state by observing the simulation and the total kinetic energy plots. The total kinetic energy becomes negligible for cases with lower magnetic cohesion (lower susceptibility and field strength), as all particles stop moving.

However, in some cases with higher magnetic cohesion, not all particles come to a complete stop. Some particles fall into a steady state around local magnetic fields, oscillating around a local minimum.

Once the assembly is settled, we use this simulation frame as our *pre-avalanche state*. Then, an avalanche is initiated by deleting (using LIGGGHTS `unfix` command) the middle partition wall. The particles are then allowed to fall freely under the influence of gravity. We run each simulation for at least 1.5 s more, after which we start checking for the steady state. Once a steady state is reached, we treat the last frame as the *post-avalanche state*.

5.4 Sensitivity analysis for magnetic time step

Table 7 shows the various time-steps used. Each simulation's time step (*dt*) is kept to 5μs. The magnetic time step (mag_t) refers to the time step used for a new magnetic force calculation. The magnetic steps value used for the `fix magnetic` is calculated as $mag_steps = \frac{mag_t}{dt}$. Neighbor time step is a similar value used to calculate how often LIGGGHTS constructs a new neighbor list.

We performed a sensitivity analysis for the magnetic time step (mag_t) to ensure that this parameter does not affect our results. MDM simulations were conducted with 6.35 mm particles, magnetic susceptibility of 40.0 (the highest in our study), and applied field strengths of 5.0 mT, 15.0 mT, and 30.0 mT, as these parameters provide the most magnetic cohesion and therefore are critical for our analysis. The three magnetic time-steps considered were 10.0 ms, 1.0 ms, and 0.1 ms while keeping all other simulation parameters and particle properties (Tables 4 & 5) the same. Fig. 10 shows the total kinetic energy at each frame for the three different magnetic time steps. Table 6 shows the critical magnetic time and hertz collision time along with the ratio of the two for each case.

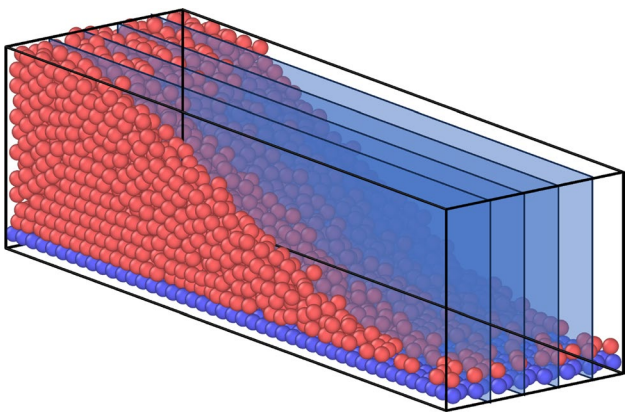


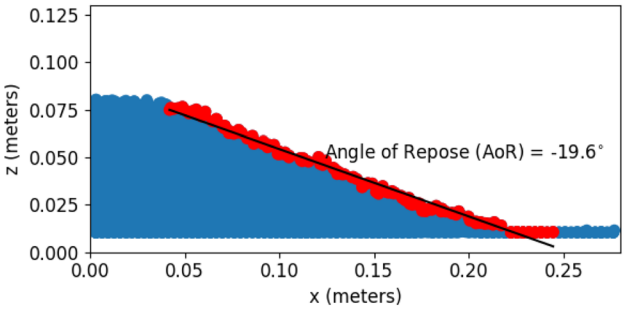
Fig. 11 Different slices to analyze the angle of repose, RMS height, and RMS slope. The simulation width is divided into five equal slices

The 10.0 ms magnetic time step never settled from a 30 mT field strength, while the 1.0 ms and 0.1 ms magnetic time steps settled for all three field strengths. The 1.0 ms and 0.1 ms magnetic time steps have similar kinetic energy profiles for all three field strengths. τ_{mag} is much higher than τ_{hc} for all three field strengths, which is expected. The order of magnitude of the ratio of τ_{mag} to τ_{hc} is around 10^3 for all three field strengths. We concluded that a magnetic time step of 1.0 ms (three orders of magnitude higher than the simulation time step of 5 μs) is suitable for our simulations, providing a good trade-off between accuracy and computational time. In all our subsequent simulations, we used a magnetic time step of 1.0 ms or 2.0 ms when particle velocities were low to speed up the simulations.

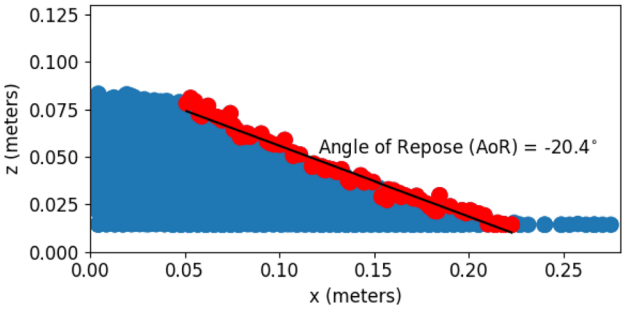
5.5 Data analysis

To analyze the simulation results, we examine three different granular flow parameters: Flow Behavior, Angle of Repose, and Surface Roughness [8]. Surface roughness is determined using the root-mean-squared (RMS) height and the RMS slope. Python, with the NumPy, SciPy, and Pandas libraries, is used for data analysis, and Ovito Pro [57] is used for visualization. We visually analyze the flow behavior to assess whether our model can recreate the three flow regimes: granular, correlated, and plastic. The angle of repose and surface roughness were analyzed by comparing the top particles in the pre-avalanche and the post-avalanche state. To find the top particles, we divide the simulation box into a grid along the *x* axis with cells of particle radius size ($a \times w \times h$, where *a* is the particle radius, *w* is the simulation width, and *h* is the height of the simulation box). We then find the highest particle in each grid cell and look at any particle with vertical displacement less than half the particle radius $\Delta y < \epsilon = a/2$ to find the list of stationary particles that haven't moved between the two frames.

We look at two different values for the angle of repose (θ) and surface roughness (RMS height ξ and RMS slope θ_{rms}). First, as in Sunday et al. [8], we collapse the $y-$dimension (width) of the simulation and find general values. Second, we also analyze the variation of these parameters in the $y-$axis (width) by calculating them for different slices (Fig. 11). We divide the simulation width into five equal *y*-slices and compute the angle of repose, RMS height, and RMS slope for each *y*-slice. Therefore, we are able to obtain multiple unique values for each of our three data analysis parameters: 3 or 4 from different seeds and 5 within each seed from the $y-$slicing.



(a) 6.35 mm particles



(b) 8.73 mm particles

Fig. 12 Final frame for seed 1 with no applied magnetic field. (a) 6.35 mm particles, (b) 8.73 mm particles

Table 8 Angle of repose and surface roughness values for both particle sizes with no applied magnetic field

a	φ	θ	ξ	θ_{rms}
(mm)	(-)	(°)	(mm)	(°)
6.35	0.62	19.97 ± 0.67	5.33 ± 0.50	23.46 ± 1.12
8.73	0.59	20.12 ± 1.02	6.53 ± 0.96	26.06 ± 1.00

5.5.1 Angle of repose

We measured the angle of repose (θ) by fitting a line through the centers of the top-most particles in the final post-avalanche frame. Some particles were removed from the left and the right side before fitting the line. Stationary particles between the pre-avalanche and post-avalanche frames are removed from fit consideration. We find the stationary particles by comparing only the $y-$displacement ($\Delta y < a/2$). The right side is trimmed to the furthest particle, resulting in the best fit (R^2 value).

5.5.2 Surface roughness

Both RMS height (ξ) and RMS slope (θ_{rms}) are calculated for all the top particles in the post-avalanche state without any trimming. The RMS height is given by

$$\xi = \left[\frac{1}{n-1}\sum_{i=1}^{n}(z_i - \bar{z})^2\right]^{1/2}, \tag{28}$$

where n is the number of particles considered, z_i is the height of the ith particle and $\bar{z}$ is the mean height of the n particles. The RMS slope is calculated by

$$\theta_{rms} = \tan^{-1}\left(\frac{v(\Delta x)}{\Delta x}\right), \tag{29}$$

where Δx is the difference in $x-$coordinate of consecutive particles and

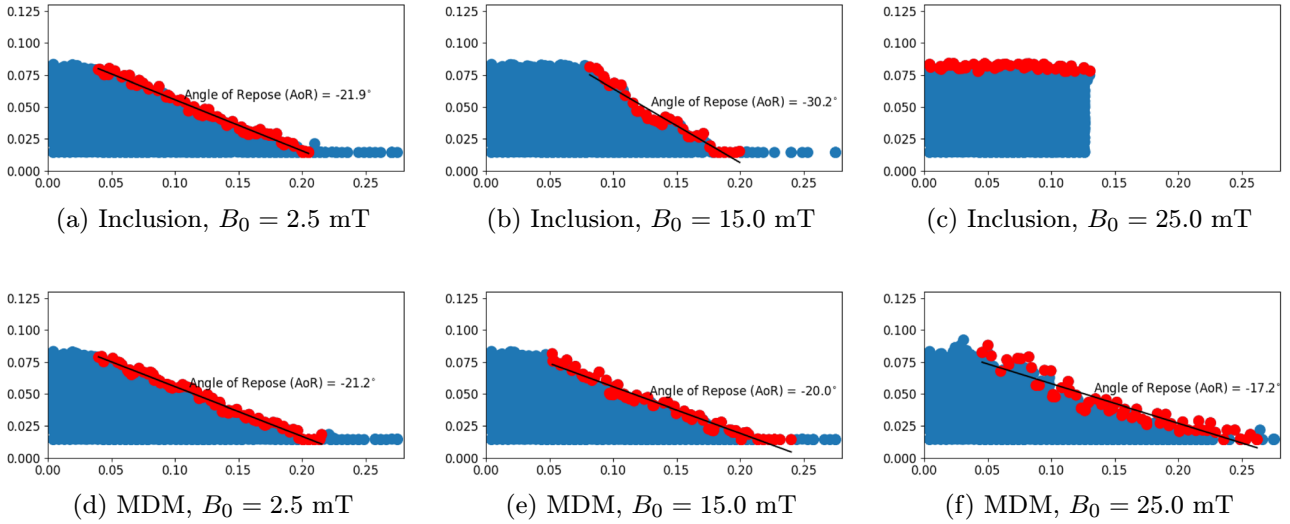
$$v(\Delta x) = \left[\frac{1}{n}\sum_{i=1}^{n}[z_i - z_{i+1}]^2\right]^{1/2}. \tag{30}$$

The difference in x-coordinate (Δx) remains constant as the particle radius (a).

5.6 Non-magnetic results

Figure 12 shows the final frames when no external magnetic field is applied. We performed non-magnetic simulations with both particle sizes (6.35 mm and 8.73 mm) to validate our interaction properties (Table 5). The fine particles (6.35 mm) were simulated first to calibrate the interaction properties by matching the angle of repose to approximately 19° [8]. Based on our four different seeds with 6.35 mm particles with a packing fraction of $\varphi = 0.62 \pm 0.01$, we calculated the angle of repose as $19.97° \pm 0.67°$. The surface roughness parameters were: RMS height $\xi = 5.33 \pm 0.50$ mm and RMS slope $\theta_{rms} = 23.46° \pm 1.12°$.

We then used the same interaction properties for the coarse particles (8.73 mm). The angle of repose was calculated as $20.12° \pm 1.02°$, which is within the experimental range of $20.15° \pm 0.17°$ [8]. The surface roughness parameters were calculated: RMS height 6.53 mm ± 0.96 mm and RMS slope $26.06° \pm 1.00°$. These values validated the calibrated interaction properties (Table 5) for both particle sizes. Table 8 summarizes the angle of repose and surface roughness values for both particle sizes.



(a) Inclusion, $B_0 = 2.5$ mT (b) Inclusion, $B_0 = 15.0$ mT (c) Inclusion, $B_0 = 25.0$ mT

(d) MDM, $B_0 = 2.5$ mT (e) MDM, $B_0 = 15.0$ mT (f) MDM, $B_0 = 25.0$ mT

Fig. 13 Magnetic Results (8.73 mm diameter and 20.0 magnetic susceptibility): Post-avalanche state for different magnetic field strengths. The top row shows the Inclusion model results, while the bottom row shows the MDM model results. Inclusion Model transitions to correlated regime at 15.0 mT (middle) before turning plastic at 25.0 mT (right) in agreement with the experiments [8], while MDM model remains in a granular state even at 25.0 mT

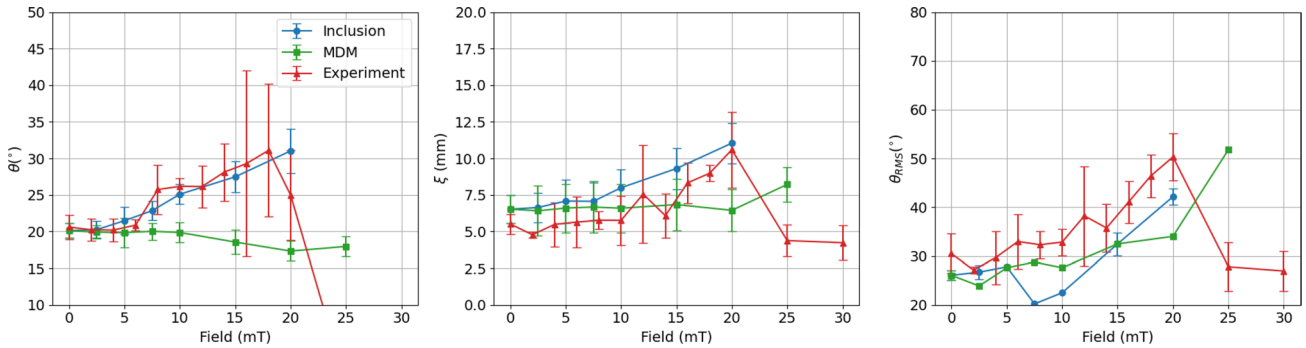
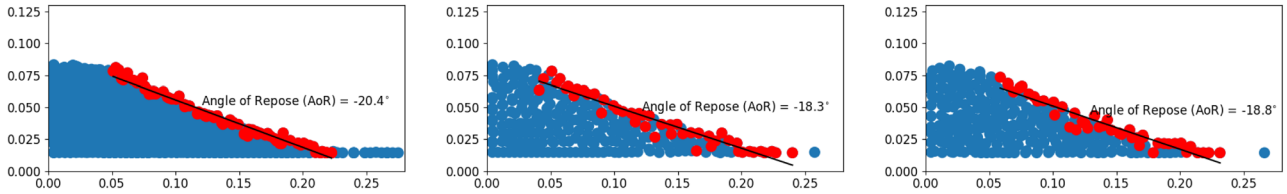


Fig. 14 Magnetic Results (8.73 mm diameter and 20.0 magnetic susceptibility): Plots showing the relationship between applied magnetic field and various parameters for both the Inclusion and MDM simulations, along with the experimental data from [8]. The Inclusion model (our implementation) and the experimental data agree on all three parameters. Surface roughness plots are shifted as compared to the experimental data. We attribute this shift to differences in data analysis techniques as discussed in Sect. 5.7

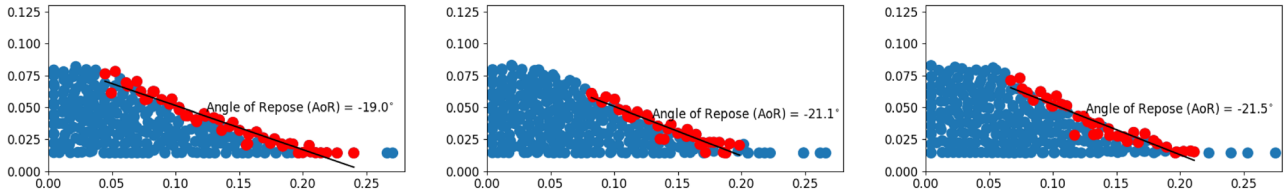
5.7 Magnetic results

We performed avalanching simulations with both the MDM and Inclusion magnetic models. The magnetic susceptibility was calibrated to 20.0 based on simulations matching the flow behavior to experiments [8] as explained in Sect. 5.2. The magnetic field was varied from 2.5 mT to 25 mT for both models.

We used three different seeds for 8.35 mm particles with the Inclusion model for each simulation to account for variability in the initial packing. Figure 13 shows the final frames for the Inclusion model with different field strengths. The results reproduced the three flow regimes: granular, correlated, and plastic at different field strengths. The correlated regime was observed at 15-20 mT, while the plastic regime was observed at 25 mT. Figure 14 shows the angle of repose and surface roughness trends with increasing field strength for MDM and the Inclusion model, along with the experimental data in Sunday et al. [8]. We demonstrate the same quantitative increase in the angle of repose and surface roughness parameters as observed in experiments [8]. The angle of repose increased from $20.12° \pm 1.02°$ (no field) to $31.03° \pm 3..00°$ (20 mT). The RMS height increased from 6.53 ± 0.96 mm (no field) to 11.04 ± 1.38 mm (15 mT). The RMS slope increased from $26.06° \pm 1.00°$ (no field) to $42.15° \pm 1.62°$ (20 mT). The surface roughness trends are shifted between the experimental and Inclusion models. We attribute this to different data analysis techniques. Sunday et al. [8] used image processing to find the profile of the top of the avalanche pile. We, on the other hand, use particle center data for analysis. This discrepancy results in shifts in the trends for surface roughness parameters, while the

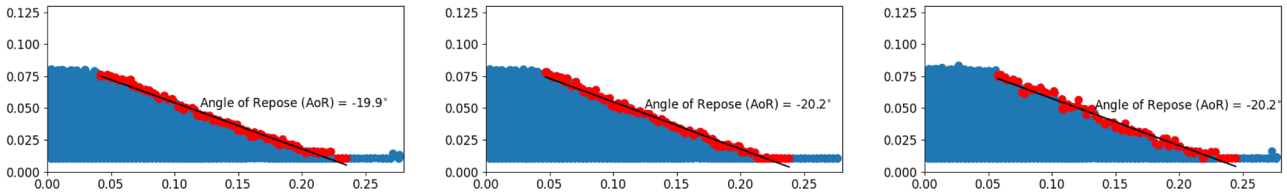


(a) No magnetic field, collapsed width (b) No magnetic field, slice 1 (c) No magnetic field, slice 2

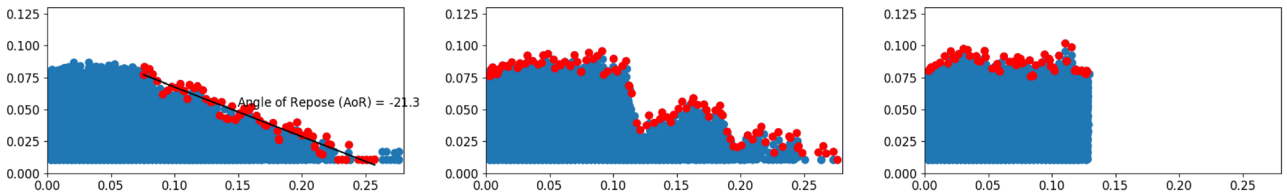


(d) No magnetic field, slice 3 (e) No magnetic field, slice 4 (f) No magnetic field, slice 5

Fig. 15 Slice Variability (8.73 mm diameter): Post-avalanche state for no applied magnetic field case with different slices



(a) $B_0 = 5.0$ mT (b) $B_0 = 15.0$ mT (c) $B_0 = 30.0$ mT



(d) $B_0 = 50.0$ mT (e) $B_0 = 65.0$ mT (f) $B_0 = 80.0$ mT

Fig. 16 Inclusion Results (6.35 mm diameter and 3.6 magnetic susceptibility): Post-avalanche state for different magnetic field strengths. Fields 5 mT to 50 mT (**a**-**d**) show a granular regime, while 65 mT (**e**) shows a correlated regime, and 80 mT (**f**) shows a plastic regime

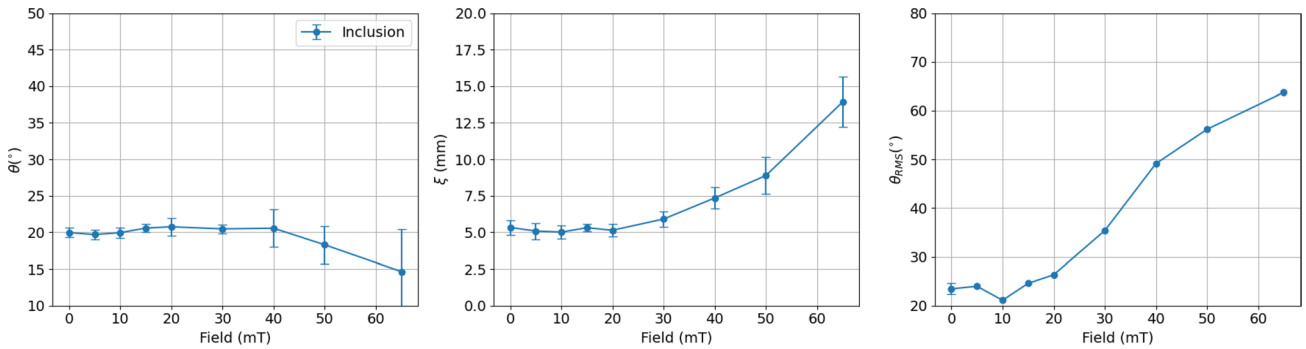


Fig. 17 Inclusion Results (6.35 mm diameter and 3.6 magnetic susceptibility): Plots showing the relationship between applied magnetic field and various parameters

percentage increase remains the same. Therefore, the modeling matches the experimental results.

We do not include the RMS slope data from the 5 slices, as the RMS slope is sensitive to the number of particles considered, and the slices significantly reduce the number of particles. The RMS slope values from the slices were all over $40°$. Figure 15 shows the top particles considered in each slice for the no field case. Each slice has a different number of top particles considered, resulting in higher RMS slope values compared to the collapsed width case.

We performed MDM simulations for a single seed to compare the results to the Inclusion model. Figure 13 shows the final frames for the MDM model with different field strengths, while Fig. 14 shows the angle of repose and surface roughness trends with increasing field strength. The figures show that the MDM model cannot reproduce the three flow regimes. We could not reach the plastic regime even at the highest field strength of 25 mT. Moreover, the MDM model continued to exhibit a granular flow regime at the fields, where the Inclusion model showed a correlated and plastic regime (15-25 mT). We observed that, at higher fields, even a single moving particle can disrupt the entire pile. This behavior led to longer settling times (> 5s) while all Inclusion simulations reached steady state within 2 s. Without the correlated and plastic regimes, the surface roughness parameters did not show a significant increase with increasing field strength. The angle of repose instead showed a slight decrease, $20.34° \pm 0.81°$ (no field) to $17.96° \pm 1.33°$ (25 mT).

We were not able to perform Inclusion simulations with 6.35 mm particles and higher susceptibilities due to computational limitations. Therefore, simulations with a single seed were performed with a magnetic susceptibility of 3.6. At the same time, we varied the field strength from 5 mT to 100 mT to achieve similar magnetic cohesion as the 8.73 mm particles with a susceptibility of 20.0 at 25 mT. We were able to reproduce the three flow regimes with the Inclusion model for 6.35 mm particles. We observed a granular flow regime at 5 mT, a correlated flow regime at 30-65 mT, and a plastic flow regime at 80-100 mT. Figure 16 shows the final frames for the Inclusion model with different field strengths, while Fig. 17 shows the angle of repose and surface roughness trends with increasing field strength. The slight increase in angle of repose from $18.81° \pm 0.17°$ (no field) to $20.50° \pm 0.20°$ (30 mT) is attributed to the lower susceptibility. While 8.73 mm simulations validated the model against experimental results, the 6.35 mm simulations showcase the three flow regimes for another particle size and susceptibility pair, providing further confidence that the Inclusion model accurately models the magnetic interactions.

6 Limitations

A few limitations of the current implementation in LIGGGHTS are worth noting. LIGGGHTS processor mapping in large simulations leads to a non-uniform distribution of particles across processors. This results in an inefficient use of processor resources and longer computation times. Changing the LIGGGHTS processor distribution or using a different DEM software like LAMMPS (`balance` command) helps achieve a more uniform distribution and therefore faster computation times. This could also enable us to run the Inclusion model with larger multipoles to achieve higher magnetic susceptibilities.

LIGGGHTS calculates forces between particle neighbor pairs only. The current implementation circumvents this limitation by using a large neighbor skin distance to include all particles as neighbors. This leads to issues with neighbor list construction requiring more computational resources. The assignment of particles to processors is carried out whenever a neighbor list is constructed, which can lead to a significant slowdown in the simulation. Future work could include optimizing the neighbor list construction to reduce the computational overhead.

A very quirky behavior was observed while using multiple `run` commands in LIGGGHTS. The `run` command in LIGGGHTS is used to advance the simulation by a specified number of time-steps. Each `run` command starts with a preconstruction of neighbor lists and forces. We observed an influx of energy into the system whenever a subsequent `run` command was used after the first one. This was not observed when using a single `run` command for the entire simulation. We suspect this is due to the large neighbor skin distance that includes all particles as neighbors. We tried using the `run pre no` command, which does not reconstruct the forces, but it did not work as expected. This issue can be avoided by using a single `run` command for the entire simulation. But this is not always feasible for longer simulations, for example, removing the middle wall requires at least two `run` commands. We therefore experimented with the `run` command and found that the influx of energy is dissipated within 1-10 dump time-steps, which is set to 1 ms in our simulations. Therefore, we can safely ignore the influx of energy if we use `run` commands with time steps larger than an order of magnitude of 10 ms.

Along with the limitations in LIGGGHTS, the Inclusion model itself has some limitations. The model is based on the assumption that all particles are identical (same magnetic susceptibility and size) and spherical. Therefore, this model cannot simulate a more complex particle shape or distribution. While the particle shape effects can be modeled by varying friction parameters, different magnetic susceptibilities cannot be modeled. We can, however, use the

MDM model to simulate different magnetic susceptibilities and particle sizes.

7 Conclusions

We have presented an accurate and experimentally validated magnetic force model for discrete element method (DEM) simulations of metallic grains. The model is based on the Inclusion model of Keaveny and Maxey [3] and is implemented in the open-source software LIGGGHTS. We also defined a new critical magnetic time to calculate the magnetic force at a lower frequency than the DEM time step. This new model is more accurate than the readily used Mutual Dipole Model (MDM), as shown in our comparisons. We validated the model against simple 1-D test cases and then used it to simulate avalanching of metallic grains. Earlier studies by Fazekas et al. [58] and Taylor et al. [11] using dipolar models did not reproduce the plastic regime. Our model replicated the bulk behavior of metallic grains in avalanching experiments by Sunday et al. [8], showcasing three distinct flow regimes: granular, correlated, and plastic. While earlier computational studies did showcase an increase in the angle of repose with an external magnetic field, they did not compare the trends with experimental results. We demonstrated a quantitative increase in the angle of repose and surface roughness parameters with increasing magnetic field strength, in agreement with the experimental data.

7.1 Future work

The current Angle of Repose (AoR) and surface roughness analysis were primarily done in a 2D manner by collapsing the $y-$dimension. We sliced the simulation width into five equal slices to analyze the variation of these parameters in the $y-$axis. Future work should include a more detailed 3D analysis of the angle of repose and surface roughness parameters. The current analysis is limited to comparing the pre-avalanche and post-avalanche states. An advantage of simulations over experimental work is the ability to track the trajectory of each particle, along with the force data. We can use this information to analyze avalanche dynamics in more detail by looking at force chains and particle interactions.

The new magnetic force model enables us to simulate more accurate and complex scenarios for magnetic cohesion in granular materials. We can use the Inclusion model to simulate the effect of particle shape on avalanching behavior by varying the rolling friction parameters. We can also model the effect of gravity on the change in angle of repose and surface roughness. These simulations would enable us to better understand the behavior of metallic regolith in microgravity environments on asteroids like 16-Psyche. The future work could also include modeling different avalanching experiments, such as the ones by Tang et al. [30]. With minor tweaks, the model can also be used to simulate magnetic manipulation of lunar regolith to design magnetism-based technologies as proposed by Kawamoto et al. [49]. Along with planetary applications, the model can also simulate metallic grains for industrial applications such as powder metallurgy and additive manufacturing.

Appendix A: Spherical harmonics approximation

We present a brief overview of the spherical harmonics approximation [3] used in the Inclusion model.

Consider a system of two particles separated by a center-to-center distance c in an applied uniform magnetic field $\mathrm{H}_0 = H_0(h_\perp \hat{x} + h_\parallel \hat{z})$. The magnetic scalar potential Φ satisfies the Laplace equation as described in equations 3-4. This problem was solved by Keaveny and Maxey [3] using a sum of spherical harmonics,

$$\Phi_{in}^{(n)} = \sum_{l=0}^{\infty} \sum_{m=0}^{1} \alpha_{lm}^{(n)} r_n^l P_l^m(\cos\theta_n) \cos m\phi_n \tag{A1}$$

$$\begin{aligned} \Phi_{out} &= -H_0\left(h_\perp x + h_\parallel z\right) , \\ &+ \sum_{n=1}^{2} \sum_{l=0}^{\infty} \sum_{m=0}^{1} \beta_{lm}^{(n)} \frac{P_l^m(\cos\theta_n)}{r_n^{(l+1)}} \cos m\phi_n , \end{aligned} \tag{A2}$$

where (x, y, z) are the Cartesian coordinates of the system and (r_n, θ_n, ϕ_n) are the spherical coordinates of a system with the origin at the center of sphere n. The associated Legendre function is defined as

$$P_l^m(\cos\theta) = (-1)^m (1 - \cos^2\theta)^{\frac{m}{2}} \frac{\mathrm{d}^m P_l(\cos\theta)}{\mathrm{d}(\cos\theta)} . \tag{A3}$$

The coefficients $\alpha_{lm}^{(n)}$ and $\beta_{lm}^{(n)}$ are found by satisfying the boundary conditions. We need the Hobson formula to describe the general solution in a single coordinate system for a pair of spheres separated by a distance c:

$$\frac{P_l^m(\cos\theta_1)}{r_1^{l+1}} = \sum_{s=m}^{\infty}(-1)^{s+m}(-1)^{l+s}\binom{l+a}{s+m}\frac{r_2^s}{c^{l+s+1}}P_s^m(\cos\theta_2)\ , \tag{A4}$$

$$\frac{P_l^m(\cos\theta_2)}{r_2^{l+1}} = \sum_{s=m}^{\infty}(-1)^{s+m}\binom{l+a}{s+m}\frac{r_1^s}{c^{l+s+1}}P_s^m(\cos\theta_1)\ . \tag{A5}$$

On substituting the general solution into the boundary conditions, using the Hobson formula and eliminating $\alpha_{lm}^{(n)}$, we find for each $m = 0, 1$,

$$\begin{aligned}&\big((\chi+2)\ l+1\big)\beta_{lm}^{(1)} + \sum_{s=1}^{\infty}\beta_{sm}^{(2)}\\ &(-1)^{l+m}\chi l\binom{l+s}{s-m}\frac{a^{2l+1}}{c^{l+s+1}}\\ &= \delta_{1j}a^3\chi H_0(\delta_{0m}h_{\parallel} - \delta_{1m}h_{\perp})\ ,\end{aligned} \tag{A6}$$

$$\begin{aligned}&\big((\chi+2)\ l+1\big)\beta_{lm}^{(2)} + \sum_{s=1}^{\infty}\beta_{sm}^{(1)}\\ &(-1)^{l+s}(-1)^{l+m}\chi l\binom{l+s}{s-m}\frac{a^{2l+1}}{c^{l+s+1}}\\ &= \delta_{1j}a^3\chi H_0(\delta_{0m}h_{\parallel} - \delta_{1m}h_{\perp})\ .\end{aligned} \tag{A7}$$

Considering only a finite number of multipoles, $l = 0, \ldots, L$, we arrive at $2L \times 2L$ linear systems of equations. For each $m = 0, 1$

$$\begin{pmatrix} \mathrm{X} & \boldsymbol{\Delta}_m \\ \boldsymbol{\Gamma}_m & \mathrm{X}\end{pmatrix}\begin{pmatrix}\beta_m^{(1)}\\ \beta_m^{(2)}\end{pmatrix} = \begin{pmatrix}\mathrm{q}_m\\ \mathrm{q}_m\end{pmatrix}\ , \tag{A8}$$

where

$$\begin{aligned}\beta_m^{(1)} &= \Big(\beta_{1m}^{(1)}, \beta_{2m}^{(1)}, \ldots, \beta_{Lm}^{(1)}\Big)^T,\\ \beta_m^{(2)} &= \Big(\beta_{1m}^{(2)}, \beta_{2m}^{(2)}, \ldots, \beta_{Lm}^{(2)}\Big)^T,\end{aligned} \tag{A9}$$

$$\begin{aligned}\mathrm{q}_0 &= \Big(H_0h_{\parallel}a^3\chi, 0, \ldots, 0\Big)^T,\\ \mathrm{q}_1 &= \Big(-H_0h_{\perp}a^3\chi, 0, \ldots, 0\Big)^T\ ,\end{aligned} \tag{A10}$$

and the $L \times L$ matrices are

$$X_{ij} = \delta_{ij}\big((\chi+2)\ i+1\big), \tag{A11}$$

$$\Delta_{ij;m} = (-1)^{i+m}\chi i\binom{i+j}{j-m}\frac{a^{2i+1}}{c^{i+j+1}}, \tag{A12}$$

$$\Gamma_{ij;m} = (-1)^{i+j}\delta_{ij;m}\ . \tag{A13}$$

After solving the system of equations, with $\beta_{lm}^{(n)}$ known, the contribution of the magnetic field from the particles is found by taking the gradient of the scalar potential Φ_{out}:

$$\begin{aligned}H_r = \sum_{l=0}^{L}\sum_{m=0}^{1}\Big[&(l+1)\beta_{lm}^{(1)}\frac{P_l^m(\cos\theta_1)}{r_1^{l+2}} - \beta_{lm}^{(2)}\sum_{s=m}^{L}\\ &(-1)^{s+m}\binom{l+s}{s+m}s\frac{r_1^{s-1}}{c^{l+s+1}}P_s^m(\cos\theta_1)\Big]\cos m\phi\ ,\end{aligned} \tag{A14}$$

$$\begin{aligned}H_\theta = -\sum_{l=0}^{L}\sum_{m=0}^{1}\Big[&\frac{\beta_{lm}^{(1)}}{r_1^{l+2}}\frac{\mathrm{d}P_l^m(\cos\theta_1)}{\mathrm{d}\theta_1} + \beta_{lm}^{(2)}\sum_{s=m}^{L}\\ &(-1)^{s+m}\binom{l+s}{s+m}\frac{r_1^{s-1}}{c^{l+s+1}}\frac{\mathrm{d}P_s^m(\cos\theta_1)}{\mathrm{d}\theta_1}\Big]\cos m\phi\ ,\end{aligned} \tag{A15}$$

$$\begin{aligned}H_\phi = \sum_{l=0}^{L}\Big[&\beta_{l1}^{(1)}\frac{P_l^1(\cos\theta_1)}{r_1^{l+2}\sin\theta_1} + \beta_{l1}^{(2)}\sum_{s=1}^{L}(-1)^{s+1}\\ &\binom{l+s}{s+m}\frac{r_1^{s-1}}{c^{l+s+1}\sin\theta_1}P_s^1(\cos\theta_1)\Big]\sin\phi\ .\end{aligned} \tag{A16}$$

The magnetic force on particle 1 is then calculated as

$$F_x = \int_{r_1=a} T_{xr}\mathrm{d}S, \tag{A17}$$

$$F_y = 0, \tag{A18}$$

$$F_z = \int_{r_1=a} T_{zr}\mathrm{d}S, \tag{A19}$$

where the stress tensor components are given by $T_{ij} = \mu_0 H_i H_j - \delta_{ij} \frac{\mu_0}{2} H_k H_k$.

We solved the dependence on ϕ analytically using Mathematica to reduce the surface integrals to polynomials in θ. First, we add the uniform magnetic field contribution and rewrite the magnetic field components as

$$H_r = P + A\cos\phi, \tag{A20}$$

$$H_\theta = Q + B\cos\phi, \tag{A21}$$

$$H_\phi = 0 + C\sin\phi, \tag{A22}$$

where P, Q, A, B, and C are functions of θ only. After substituting these expressions into the stress tensor components, we find the x and z force integrals reduced to:

$$F_x = \int_0^\pi \pi \Big[-CP + (BP + AQ)\cos\theta + (AP - BQ)\sin\theta \Big] \mathrm{d}\theta, \tag{A23}$$

$$F_z = \int_0^\pi \frac{\pi}{2} \Big[(A^2 - B^2 - C^2 + 2P^2 - 2Q^2)\cos\theta - 2(AB + 2PQ)\sin\theta \Big] \mathrm{d}\theta \ . \tag{A24}$$

Finally, we used Gaussian quadrature to evaluate the integrals over θ.

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Author contributions AS and CMH conceptualized this study together. AS developed the model; implemented, carried out, and analyzed the simulation results in this paper, and was also responsible for original manuscript preparation. CMH obtained funding, analyzed simulation results, and provided manuscript editing and review.

Data availability Given the size of our datasets, we provide a subset of data used in this paper. The restart files and scripts needed to launch avalanching simulations are provided in [56]. We also provide the two dump files used to analyze the Angle of Repose and the Surface Roughness parameters. Any reader can use the provided files with open-source software [52] to easily recreate our work and validate it.

Declarations

Conflict of interest The authors declare no Conflict of interest.

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